Multi-Class Annotation Errors in Causal Inference: Topology-Dependent Bias, Sensitivity, and Correction

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Abstract

Large language models are increasingly used as annotators in computational social science, yet annotation errors propagate into downstream causal estimates in ways that depend on both the confusion matrix structure and the causal DAG topology. We derive bias expressions for seven canonical topologies under K-class non-differential misclassification and show that $K \geq 3$ introduces a structural possibility of sign reversal absent in the binary case. We prove sign preservation for confounding and collider, and verify it numerically for mediation, M-bias, and front-door; exposure and IV topologies can reverse the treatment effect sign. Under realistic LLM accuracy (diagonal ≥ 0.7), the sign reversal rate is 2.1%, exclusively in secondary coefficients—in all 23 empirical configurations tested, the primary treatment effect does not reverse—consistent with the largest-magnitude coefficient dominating the misclassification-induced weighted sum. We introduce Annotation Sensitivity Values (ASVs) revealing a vulnerability spectrum from IV (maximally fragile) to M-bias (immune), and show that matrix-based corrections are provably ineffective for four topologies due to Frisch–Waugh–Lovell invariance. Topology-aware analysis is necessary for valid causal inference with LLM-generated annotations.

1 Introduction

Large language models are rapidly replacing human annotators in computational social science, with researchers using LLM-generated labels for tasks ranging from sentiment classification to political stance detection (Gilardi et al., 2023; Ziems et al., 2024; Egami et al., 2023; Baumann et al., 2025). Crucially, most of these tasks are inherently multi-class: stance detection uses 3–4 categories, sentiment analysis 5, and disaster response taxonomies 10 or more. This shift promises dramatic cost reduction, but introduces a critical validity gap: annotation errors from imperfect classifiers propagate into downstream causal estimates in ways that depend on both the error structure and the causal design. Baumann et al. (2025) demonstrate that across 37 annotation tasks, 31% of hypotheses yield incorrect conclusions with state-of-the-art LLMs, rising to 50% for smaller models; these errors are highly correlated across models (60% agreement when both err) (Kim et al., 2025), undermining ensemble-based mitigation. Analyzing the *structure* of annotation error, rather than simply reducing its magnitude, is essential for valid downstream inference.

The transition at $K \geq 3$ is not gradual but structural. For $K=2$, the stochastic constraint forces any confusion matrix with positive diagonal to preserve coefficient signs—an algebraic property of the 2×2 matrix dimension (Brenner, 1993). This guarantee *vanishes* at $K \geq 3$: the higher-dimensional confusion matrix admits negative effective weights on categories with opposite-sign coefficients, enabling sign reversal even under high accuracy (Veierød and Laake, 2001). Under a uniform Dirichlet prior, 83.4% of $K=3$ exposure confusion matrices produce sign reversal of at least one secondary coefficient; under realistic accuracy (diagonal $\geq$ 0.7), 2.1% still do. The primary treatment coefficient, however, is never reversed—a structural property we explain in Section 3.

Standard classification metrics do not predict downstream causal risk: within a narrow F_1 range of 0.614–0.690 across three frontier LLMs, relative bias varies by an order of magnitude across topologies (from 4.1% to 42.1%), illustrating that *topology*, not annota-

tor quality, determines downstream risk. This fundamental asymmetry has received limited systematic treatment. Dosemeci et al. (1990) and Veierød and Laake (2001) noted the possibility of polytomous sign reversal; Brenner (1993) identified amplification for polytomous confounders; Lee and Wood-Doughty (2024) extended classifier error analysis to categorical variables and LLM classifiers, but covered only one DAG topology. No prior work systematically characterizes how the *same* confusion matrix produces qualitatively different bias across causal designs.

We address these gaps with three contributions:

1. **Multi-class bias taxonomy.** We derive bias expressions for seven canonical DAG topologies (Figure 1) under K-class non-differential misclassification, prove sign preservation for confounding and collider, verify it numerically for mediation, M-bias, and front-door, and show that exposure and IV can produce sign reversal. Binarization amplifies the gap: collapsing K=3 to binary introduces additional bias varying 154× across topologies.

2. **Topology-aware correction guidance.** We evaluate five correction methods and prove a Frisch–Waugh–Lovell invariance result: matrix-based corrections achieve exactly 0% bias reduction for four topologies where A enters as a covariate—a negative result overlooked in the annotation correction literature. PPI++ achieves 47.0% RMSE reduction but requires gold labels and is harmful for M-bias (−88%).

3. **Annotation Sensitivity Values.** As a corollary of the bias taxonomy, we define three complementary ASV types—magnitude, significance, and sign-flip—revealing a topology-dependent vulnerability spectrum from IV (maximally fragile, 52.6% sign reversal) to M-bias (immune, ASV = ∞).

2 Related Work

Measurement error in causal inference. The foundations are established in Carroll et al. (2006), Gustafson (2004), Buonaccorsi (2010), and Bound et al. (2001), with starred-node DAGs formalized by Hernán and Cole (2009) and DAG-based correction theory developed by Kuroki and Pearl (2014). For binary variables, non-differential misclassification biases toward the null (Brenner, 1993; Ogburn and VanderWeele, 2012), but this fails for polytomous exposures (Dosemeci et al., 1990; Jurek et al., 2005; Veierød and Laake, 2001). Edwards et al. (2021) showed dependent misclassification can bias in either direction, a concern that also drives mediation analyses with mismeasured mediators (Valeri and VanderWeele, 2013). All analyze a single DAG topology; ours is the first to characterize how the *same* confusion matrix produces qualitatively different bias across seven topologies.

Text classifiers and LLM annotation. Grimmer et al. (2022) and Feder et al. (2022) provide foundations for text-as-data and causal inference in NLP. Wood-Doughty et al. (2018) first applied DAG reasoning to classifier error; TeBlunthuis et al. (2024) gave the most complete binary treatment (K=2, four DAGs); Lee and Wood-Doughty (2024) extended to categorical LLM classifiers but covered only confounding. On annotation quality, Gilardi et al. (2023) and Ziems et al. (2024) document the rapid adoption of LLM annotators, Baumann et al. (2025) found 31% incorrect conclusions, Pangakis et al. (2023) demonstrated that automated annotations require task-specific validation, Cheng et al. (2024) propose structured approaches to reduce LLM measurement error, and Kim et al. (2025) showed 60% error correlation across models. Audinet de Pieuchon et al. (2025) benchmark debiasing; Keith et al. (2020) survey the broader landscape. These works either restrict to binary labels, cover a single topology, or diagnose quality without connecting it to causal validity.

Sensitivity analysis and correction methods. Our ASV framework draws on E-values (VanderWeele and Ding, 2017), the robustness value of Cinelli and Hazlett (2020), and the mediation sensitivity analysis of Imai et al. (2010), but parametrizes annotation error severity rather than omitted variable strength or sequential ignorability violations. The quantitative bias analysis

framework of Lash et al. (2009) provides a complementary epidemiologic perspective, and Mahajan (2006) establishes identification of regression models with misclassified regressors via instrumental variables. The unified measurement-error and missing-data framework of Blackwell et al. (2017) similarly motivates treating annotation noise as a first-class object of inference. For correction, PPI++ (Angelopoulos et al., 2023) and DSL (Egami et al., 2023) are topology-agnostic; Zrnić and Candès (2024) extend PPI to gold-free settings, and Gligorić et al. (2025) leverage LLM confidence scores. Our contribution reveals that this agnosticism is costly: correction effectiveness is itself topology-dependent, with matrix-based methods provably ineffective for four topologies. Bayesian measurement error models (Richardson and Gilks, 1993; Gustafson, 2004) and EM imputation offer alternatives that bypass FWL invariance.

3 Method

3.1 Setup and Notation

Let A denote a true K-class categorical variable (e.g., sentiment with $K{=}3$ levels) and A^* its observed, potentially misclassified version produced by an LLM annotator. The misclassification process is characterized by a $K{\times}K$ confusion matrix C with entries $C_{jk} = P(A^*{=}j \mid A{=}k)$, where each column sums to one. Following Hernán and Cole (2009), we represent this measurement process using *starred-node* DAGs: the true variable A is latent, and the observed A^* is connected to A via a dashed measurement edge.

Definition 1 (Non-differential misclassification)**.** *Misclassification of A is non-differential with respect to outcome Y and covariates $\mathbf{Z}$ if $P(A^*{=}j \mid A{=}k, Y, \mathbf{Z}) = P(A^*{=}j \mid A{=}k) = C_{jk}$ for all j, k.*

This assumption is natural when the LLM annotates treatment labels from source documents without access to outcome information. However, in many NLP annotation pipelines, the same text that determines A also carries information about Y—a "text as common cause" structure that creates a channel for differential misclassification even without explicit access to Y. Specific violation pathways include: (i) outcome-conditional prompting, (ii) few-shot demonstrations encoding A–Y associations, and (iii) retrieval-augmented pipelines surfacing Y-correlated documents. Practitioners can diagnose potential violations by testing $P(A^*|A, Y) \approx P(A^*|A)$ on a gold-labeled subset (see Limitations).

We study seven canonical DAG topologies (Figure 1) (Pearl, 2009) in which A occupies structurally distinct causal roles: confounder, exposure, mediator, collider, M-bias variable, front-door mediator, and instrument. Each role induces a different functional relationship between C and the probability limit of the standard causal estimator, yielding topology-dependent bias.

3.2 Multi-Class Bias Expressions

For each topology, we derive the probability limit of the naive estimator that substitutes A^* for A. Let τ denote the target causal parameter—its precise definition is topology-specific (e.g., the coefficient on X for confounding, the category-specific β_j for exposure, or the direct effect τ_{direct} for mediation)—and $\hat{\tau}(A^*)$ the estimator using misclassified annotations.

Confounding (A = confounder). When A confounds the $X{\to}Y$ relationship, the OLS estimator controlling for A^* has probability limit

$$\text{plim}\ \hat{\tau}(A^*) = \big(\mathbb{E}[Z'Z]\big)^{-1}\mathbb{E}[Z'X]\,\boldsymbol{\beta}, \qquad (1)$$

where $Z = [X, \text{dummies}(A^*)]$, $\boldsymbol{\beta}$ contains the true coefficients, and the target parameter τ is the coefficient on X. Misclassification produces residual confounding that monotonically approaches the omitted variable bias, but the sign of $\hat{\tau}$ is preserved: $\tau_{\text{sign_flip}} = 0$ structurally.[1]

Exposure (A = treatment variable). When A is the exposure of interest, the biased

[1] This result is consistent with Brenner (1993), who showed that the adjusted estimate with misclassified polytomous confounders can be biased toward or away from the null. Both findings hold simultaneously: misclassification degrades confounding control, producing residual bias whose magnitude can exceed the correctly adjusted estimate (amplification), but the treatment effect sign is never reversed. The distinction is between *sign reversal* ($\text{sign}(\hat{\tau}) \neq \text{sign}(\tau)$, which we show never occurs) and *amplification* ($|\hat{\tau}| > |\tau|$, which occurs universally in this topology).

coefficient for category j becomes a weighted combination of true coefficients:

$$\text{plim}\ \hat{\beta}_j(A^*) = \sum_{k=1}^{K} w_{jk}\,\beta_k, \tag{2}$$

where w_{jk} depends on the confusion matrix C and the marginal distribution of A. Unlike the binary case, where non-differential misclassification guarantees attenuation toward the null (Brenner, 1993), the $K{\geq}3$ case permits w_{jk} to assign negative weight to categories with opposite-sign coefficients, producing *sign reversal*. A near-permutation confusion matrix effectively swaps category labels, making sign reversal the dominant outcome (83.4% under uniform Dirichlet prior). This failure mode is structurally impossible at $K{=}2$ because a $2{\times}2$ stochastic matrix with positive diagonal cannot reverse the sign of a single coefficient.[2]

Mediation (A = mediator). When A mediates the $X{\to}Y$ relationship, the direct effect estimator controlling for A^* has probability limit

$$\text{plim}\ \hat{\tau}_{\text{direct}}(A^*) = \tau_{\text{direct}} + \big(1 - \lambda(C)\big)\,\tau_{\text{indirect}}, \tag{3}$$

where $\lambda(C) \in [0,1]$ under proportional effects (Appendix A, Proposition 4), with $\lambda(I) = 1$ under perfect classification. Misclassification ($\lambda < 1$) causes the residual indirect effect to be absorbed into the direct path estimate, producing upward bias in $|\hat{\tau}_{\text{direct}}|$ but preserving the sign ($\tau_{\text{sign_flip}} = 0$).

Collider (A = collider, conditioned on A^*). Conditioning on a collider A of X and Y opens a spurious path. When we condition on A^* instead, misclassification attenuates this spurious association because A^* is a noisy proxy for A. The collider topology thus exhibits a *protective* property: annotation error reduces rather than amplifies the bias from collider stratification, with $\tau_{\text{sign_flip}} = 0$.

[2]Empirically, the primary coefficient (largest $|\beta_k|$) never reverses across our 23 LLM configurations. This is consistent with magnitude dominance: the diagonal weight w_{jj} contributes $w_{jj}\beta_j$ to $\hat{\beta}_j(A^*)$, and when $|\beta_j|$ is sufficiently large relative to the off-diagonal contamination $\sum_{k\neq j} w_{jk}\beta_k$, the sign is preserved despite w_{jk} potentially being negative.

M-bias (A = intermediate collider). In the M-shaped DAG ($U_1 \to X$, $U_1 \to A$, $U_2 \to A$, $U_2 \to Y$, $X \to Y$), adjusting for A introduces a small spurious association via the M-bias path (Greenland, 2003). Misclassifying A weakens this adjustment, reducing the M-bias. This topology is *immune*: $\hat{\tau}(A^*)$ is closer to τ than $\hat{\tau}(A)$ for all non-trivial confusion matrices, yielding ASV $= \infty$.

Front-door (A = front-door mediator). In the front-door DAG ($X \to A \to Y$ with unmeasured $U \to X$, $U \to Y$), the front-door estimator (Bellemare et al., 2024) uses A to identify effects despite unmeasured confounding. Misclassification attenuates both $X \to A$ and $A \to Y$ relationships, producing moderate bias with $\tau_{\text{sign_flip}} = 0$.

Instrumental variable (A = instrument). The Wald estimator (Angrist and Imbens, 1995) $\hat{\tau}_{\text{IV}} = \hat{\text{cov}}(A^*, Y)/\hat{\text{cov}}(A^*, X)$ has C in the denominator. Misclassification can drive first-stage coefficients toward zero or reverse their sign, causing the ratio estimator to diverge or flip. Under random Dirichlet matrices, 52.6% produce sign reversal, making IV the most fragile topology.

Structural sign-flip guarantee.

Proposition 1 (Sign preservation—analytic results)**.** *Under non-differential misclassification with $C_{kk} > 0$ for all k:* (a) *Confounding preserves sign when* $\text{sign}(\tau) = \text{sign}(\tau{+}B_{\text{OV}})$, *where* $B_{\text{OV}}{=}\,\text{Cov}(g(A), f(A))/\,\text{Var}(T)$ *is the omitted-variable bias (Cauchy–Schwarz; Appendix A.3).* (b) *Collider preserves sign when* $|\tau| > \sqrt{\text{Var}(g)\,\text{Var}(h)}/\mathbb{E}[\text{Var}(T|A)]$ *(Berkson attenuation; Appendix A.6).* (c) *Exposure can produce sign reversal for $K{\geq}3$, even with $C_{kk}{>}0.5$ (counterexample in Appendix A.8).*

Our theoretical results establish a three-tier hierarchy: (i) analytically proved (Proposition 1(a,b), Theorem 1), (ii) numerically verified with zero violations across 100,000 matrices (mediation, M-bias, front-door sign preservation), and (iii) conjectured with ε-approximate bounds (Conjecture 1).

Numerical Verification 1 (Mediation, M-bias, front-door). Sign preservation holds under topology-specific conditions: mediation requires $\text{sign}(\tau_{\text{dir}}){=}\,\text{sign}(\tau_{\text{total}})$; front-door re-

quires $\text{sign}(\lambda\alpha_U) = \text{sign}(\lambda\alpha_U + \text{Cov}(T, f(A)))$; M-bias is attenuating for all C. Verified across 100,000 random matrices with zero violations (Appendix A.9); analytic proofs for general K remain open.

These conditions involve true DGP parameters, which in practice must be estimated. The ASV framework (§3.3) provides a complementary assessment that does not require knowledge of the exact sufficient conditions.

3.3 Annotation Sensitivity Values

Our bias taxonomy and FWL invariance results naturally motivate a practical question: how much annotation error can a study tolerate before conclusions change? We define Annotation Sensitivity Values (ASVs) for this purpose, drawing on the E-value framework (VanderWeele and Ding, 2017) but with three key differences: (1) E-values parametrize a scalar Γ that is naturally monotone; ASVs map a $K \times K$ matrix to a 1D path $C(\delta)$, requiring proof of ε-approximate monotonicity (Theorem 1); (2) three ASV types (magnitude, significance, sign-flip) address topology-dependent failure modes; and (3) ASVs combine with FWL invariance to route practitioners to topology-appropriate corrections.

Parameterization. We parameterize the confusion matrix as $C(\delta) = (1-\delta)I + \delta C_0$, where C_0 is a reference confusion matrix (e.g., estimated from a validation set) and $\delta \geq 0$ controls the severity of misclassification. At δ=0, classification is perfect; at δ=1, classification matches C_0. Non-negativity of $C(\delta)$ requires $\delta \leq \delta_{\max} = \min_j \frac{1}{1-(C_0)_{jj}}$. Since $(C_0)_{jj} \in [0, 1]$, we have $\delta_{\max} \geq 1$ for all valid C_0, ensuring the full range $[0, 1]$ is always feasible; all ASV computations are constrained to $[0, \delta_{\max}]$.

Three ASV types. We define three complementary measures: (1) **Magnitude ASV** $\delta^*_{\text{mag}} = \min\{\delta : |\text{plim } \hat{\tau}(C(\delta)) - \tau| > \epsilon\}$—analogous to E-values; (2) **Significance ASV** $\delta^*_{\text{sig}} = \min\{\delta : |\text{plim } \hat{\tau}(C(\delta))| < z_{\alpha/2} \cdot SE(N)\}$—relevant for attenuating topologies; (3) **Sign-flip ASV** $\delta^*_{\text{sf}} = \min\{\delta : \text{sign}(\hat{\tau}) \neq \text{sign}(\tau)\}$—exposure and IV only. The significance ASV depends on N through $SE(N)$, exhibiting a sharp phase transition at a critical N^* below which all rejections are blocked.

Theorem 1 (Envelope monotonicity—confounding and mediation)**.** *For confounding and mediation topologies with proportional effects ($f = \lambda g$ for some scalar λ) under $C(\delta) = (1-\delta)I + \delta C_0$ with $(C_0)_{jj} \geq \frac{1}{K}$ for all j, $|B(C(\delta))|$ is monotonically non-decreasing in $\delta \in [0, 1]$. For K=2, the proof is exact; for $K \geq 3$, worst-case deviations are < 0.005 across 20,000 random (C_0, p, g) triples per K.*

Proof sketch. Both topologies reduce to $B(C(\delta)) = \lambda V_W(\delta)/(V_W(\delta) + \sigma^2)$ where $V_W(\delta) = \text{Var}(g) - V_B(\delta)$, $V_B(\delta) = \text{Var}(\mathbb{E}[g|A^*_\delta])$, and σ^2 is a topology-specific constant (σ^2_T for confounding, $\mathbb{E}[\text{Var}(T|A)]$ for mediation). The key quantity $V_B(\delta) = \sum_j N_j(\delta)^2/q_j(\delta) - \bar{g}^2$ is a sum of perspective functions of x^2, hence *convex* in δ. Its derivative satisfies $V'_B(0) = -\sum_{j \neq k} (C_0)_{jk}\, p_k\, (g_j - g_k)^2 \leq 0$ (always). For K=2 with $(C_0)_{jj} \geq \frac{1}{2}$, we prove $V'_B(1) \leq 0$ analytically; convexity then implies $V'_B(\delta) \leq 0$ for all $\delta \in [0, 1]$. Since $V_W = \text{Var}(g) - V_B$ is non-decreasing and $x/(x+c)$ is increasing for $c > 0$, $|B|$ is non-decreasing (full proofs in Appendix A.4 and A.5). The condition $(C_0)_{jj} \geq \frac{1}{K}$ excludes degenerate confusion matrices (e.g., permutation matrices where labels are swapped rather than noised); all practical LLM annotation pipelines satisfy this.

Conjecture 1 (ε-approximate envelope monotonicity)**.** *For non-IV topologies under $C(\delta) = (1-\delta)I + \delta C_0$: collider and M-bias (protective topologies) are ε-approximately non-increasing in δ, with $\varepsilon < 0.03$ across all tested τ. Front-door is ε-approximately non-decreasing, with $\varepsilon < 0.06$ for $\tau \leq 2.0$; at extreme τ, normalization artifacts can increase ε to ≈ 0.14 (Appendix B.11). For exposure, $\varepsilon < 0.008$ at the 99th percentile, though extreme-tail matrices ($<$0.02% of Dirichlet draws) can reach $\varepsilon \approx 0.15$. Mediation envelope monotonicity is proved for proportional effects (Theorem 1) and verified numerically for the general case with $\varepsilon < 0.02$ at default DGP parameters; small-τ sensitivity is documented in Appendix B.11. Verified across 30,000 random matrices per topology, $K \in \{3, 5, 7\}$, and $\tau \in \{0.1, 0.5, 1.0, 2.0, 5.0\}$ using analytic probability limits.*

The IV topology is excluded from Conjecture 1: the Wald estimator exhibits singu-

larities when the first-stage coefficient passes through zero, producing unbounded deviations. Magnitude ASV remains well-defined away from these singularities.

When strict monotonicity holds only approximately, the reported δ^* may differ from the true threshold by at most ε/η in δ-units—bounded and practically small at the recommended $\eta = 10\%$ (e.g., ε=0.03 yields $\leq$0.3 δ-units for collider/M-bias; ε=0.14 at extreme τ yields $\leq$1.4 δ-units for front-door). ASV is most reliable for $|\tau| \geq 0.5$ in standardized units; at smaller effect sizes, practitioners should consult the DGP sensitivity analysis (Appendix B.11). Topology rankings are preserved across all tested configurations.

Binarization of K=3 annotations introduces additional topology-dependent bias (Appendix B.1).

4 Experiments

We validate the bias taxonomy and ASV framework through analytical derivations and Monte Carlo simulation across seven DAG topologies under K=3 non-differential misclassification.

4.1 Setup

For each of the seven topologies (confounding, exposure, mediation, collider, M-bias, front-door, IV), we sample 100,000 confusion matrices from a Dirichlet$(1, 1, 1)$ prior, compute the analytical probability limit of the causal estimator, and validate against Monte Carlo simulation (N=10,000). ASV experiments use $C(\delta) = (1-\delta)I + \delta C_0$.

4.2 Bias Taxonomy

Theory-MC agreement. Analytical probability limits match Monte Carlo estimates with grand MAPE of 2.0% across 126 scenarios (18 LLM configurations $\times$ 7 topologies, N=10,000). Per-topology MAPE ranges from 0.5% (confounding) to 5.5% (IV); the IV maximum reflects the Wald estimator's small-denominator variance amplification.

Exposure DAG: sign reversal under multi-class misclassification. Under the full Dirichlet$(1, 1, 1)$ prior, 85% of K=3 confusion matrices produce bias that violates the binary "toward-null" guarantee. More critically, 83.4% of matrices cause *sign reversal* of at least one coefficient. The two failure modes are distinct: for a coefficient with true value β_1=1.0, a near-permutation C produces $\hat{\gamma}_1$=$-$0.84 (sign reversal, magnitude 0.84$\times$); for β_2=$-$0.5, the same C yields $\hat{\gamma}_2$=$-$1.41 (same-sign amplification, 2.83$\times$).

This finding persists in practically relevant subspaces. When restricting to confusion matrices with diagonal entries ≥ 0.7 (typical of current LLM classifiers achieving $F_1 > 0.8$), the violation rate remains 13.7% with 2.1% sign reversal and 11.7% amplification. Even at diagonal ≥ 0.9, violation persists at 11.9%. In contrast, the K=2 control confirms that all six tested binary confusion matrices maintain the classical bias-toward-null guarantee (Brenner, 1993), establishing that the multi-class failure is structurally distinct from the binary case.

Structural guarantee for non-exposure topologies. Five of seven topologies exhibit zero sign-flip rate across all sampled matrices (Table 1). Confounding, mediation, and front-door show 100% violation and amplification but never reverse the sign.[3] Collider misclassification is protective: conditioning on noisy A^* induces *less* collider bias than conditioning on A ($|B(C)|$ approximately non-increasing; Conjecture 1). M-bias shows the weakest sensitivity (53.2% violation, 10.4% amplification).

IV: Wald estimator instability. The IV topology is the sole non-exposure exception: 52.6% of random matrices cause the Wald estimator's denominator to change sign. This rate is stable across K=3, 5, 7, confirming structural fragility. Unlike exposure, IV violation remains at 100% even at diagonal ≥ 0.9 (Appendix B.4).

All structural patterns persist at K=5, 7 (Appendix B.8).

The bias distributions are visualized in Figure 2.

4.3 ASV Framework Validation

Envelope monotonicity. The ASV envelope (bias magnitude as a function of δ) is ε-

[3] Consistent with Brenner (1993) and Dosemeci et al. (1990): polytomous misclassification can bias *away* from the null (amplification), but our sign-flip metric confirms the effect sign is preserved.

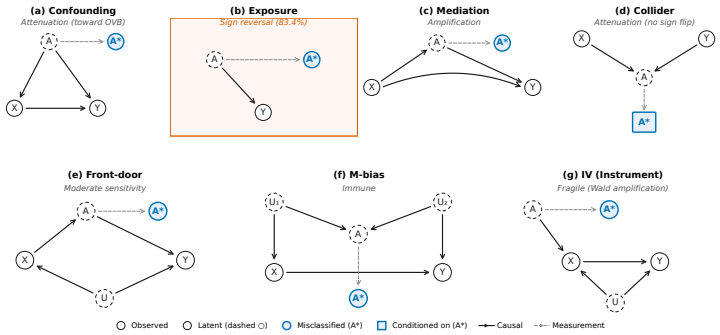


Figure 1: Seven canonical DAG topologies with starred-node annotation. A (latent) connects to A^* (observed) via a dashed measurement edge; the causal role of A varies across topologies.

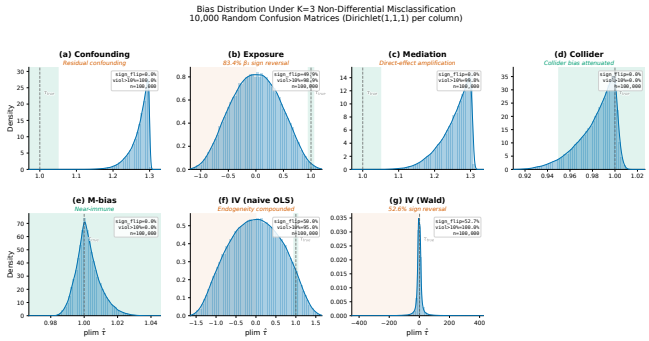


Figure 2: Bias distributions under 100,000 random K=3 confusion matrices. Red = sign reversal. Exposure: 83.4% sign reversal; IV: 52.6%.

Table 1: Bias characteristics of K=3 non-differential misclassification across 7 DAG topologies. Violation: $|\hat{\tau}| > |\tau|$; Sign flip: $\text{sign}(\hat{\tau}) \neq \text{sign}(\tau)$; Amplification: $|\hat{\tau}| > |\tau|$ with same sign; τ_{sf}=0?: sign-flip guarantee holds unconditionally (✓), under DGP-specific sufficient conditions (✓*; see Proposition 1 and Numerical Verification 1), or does not hold (✗). Rates over 100,000 Dirichlet$(1,1,1)$ random confusion matrices.

Topology	Violation	Sign Flip	Amplif.	τ_{sf}=0?
Confounding	100%	0%	100%	✓*
Exposure	85%	83.4%	14.1%	✗
Mediation	100%	0%	100%	✓*
Front-door	100%	0%	100%	✓*
Collider	100%	0%	100%	✓*
M-bias	53.2%	0%	10.4%	✓
IV	100%	52.6%	100%	✗

approximately monotonic for non-IV topologies: collider and M-bias achieve $\varepsilon < 0.03$ across all tested configurations; front-door achieves $\varepsilon < 0.06$ for moderate τ; exposure satisfies $\varepsilon_{p99} < 0.008$ (Conjecture 1; Appendix B.11). IV is excluded: the Wald estimator exhibits singularities when the first-stage coefficient passes through zero, producing unbounded deviations. These bounded deviations for non-IV topologies confirm that δ serves as a meaningful "dial" controlling annotation error severity.

Magnitude ASV. Table 3 reports magnitude ASV ranges at τ=0.5 with a 10% bias threshold, ranking from most fragile to most robust: front-door (0.064–0.387) < mediation < confounding < IV (0.101–0.607) < collider (0.338–2.028) < M-bias (∞). M-bias achieves infinite ASV because annotation error only reduces the already-small spurious path.

Significance and sign-flip ASV. At τ=0.05, significance ASV achieves 100% power for confounding, front-door, and M-bias, but 0% for mediation, collider, and IV (correct behavior for attenuating topologies; Type I error is 0%). For sign-flip ASV, 83.4% of exposure reference matrices yield $\tau_{\text{sign_flip}} < 1$ (extreme sensitivity); IV sign-flip ASV ranges 1.1–6.7 (moderate-to-high). The complete ranking (Figure 4, Table 3, Appendix B) confirms IV as maximally fragile and M-bias as immune (ASV = ∞).

***N*-sensitivity.** Significance ASV exhibits a sharp phase transition at a critical $N^* \approx$ 6,000–10,000 (analytically computable per topology), below which all rejections are blocked. Bootstrap CIs (B=1,000; Appendix B.7): $\delta^* = 0.94\,[0.60, 1.36]$ (exposure), $0.75\,[0.44, 1.29]$ (IV).

4.4 Empirical Validation

We validate using real LLM confusion matrices from three tasks: HumAID (Alam et al., 2021) (K=10), VAST (K=3), and CivilComments (K=2). Four LLMs (Llama-4-Maverick, Qwen3-235B, DeepSeek-Chat, Qwen2.5-7B-Instruct) produce 23 configurations across the three datasets under zero-shot and few-shot prompting, yielding 161 bias scenarios (23×7 topologies).

HumAID (K=10): theory–practice alignment. The empirical bias ranking matches predictions: IV (12.4–42.1%), exposure (9.0–17.4%), mediation/front-door (10.7–14.8%), while collider, confounding, and M-bias remain below 10%—consistent with the ASV vulnerability spectrum.

Cross-K comparison. Bias varies substantially across settings: VAST (K=3) produces 62.8% mean bias with IV reaching 326%, versus 12.2% for HumAID (K=10) and 32.4% for CivilComments (K=2). Even CivilComments configurations with F_1=0.82 (Qwen3-235B) produce 62–92% IV bias, confirming that classification accuracy does not predict downstream causal risk (Figure 3).[4]

Checking all $K-1$ exposure coefficients, 14 of 23 configurations (61%) exhibit sign reversal in at least one coefficient—all in the exposure topology. The primary coefficient (β_1, largest magnitude) never reverses; sign flips occur exclusively in secondary coefficients with small $|\beta|$ corresponding to rare categories. Empirically, sign reversal occurs only when treatment effects have opposite signs across categories; in all same-sign β_k configurations tested, only attenuation is observed. Zero sign reversals occur across 138 non-exposure scenarios (23×6 topologies), validating the structural guarantee K-invariantly.

ASV diagnostics flag 55.4% of HumAID and 100% of CivilComments scenarios as magnitude-unsafe despite F_1 up to 0.82 (Appendix B.9). Sign-flip patterns are preserved under logistic DGP ($|\Delta\,\mathrm{SF}_\tau| < 0.02$; Appendix B.6).

[4]VAST's near-degenerate confusion matrices (Neutral-class recall 3.6–16.1%) drive its extreme IV bias via the Wald estimator's small-denominator amplification.

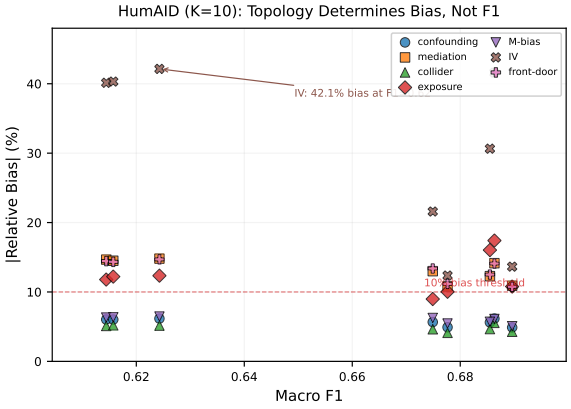


Figure 3: Topology determines downstream causal bias, not F_1. HumAID K=10 (56 points): within F_1 range 0.614–0.690, bias varies 10× (IV 42.1% vs. collider <5.6%).

4.5 Correction Guidance

We evaluate five correction methods (naive plug-in, MLA, DSL (Egami et al., 2023), MC-SIMEX, PPI++ (Angelopoulos et al., 2023)) across all seven topologies under six error structures (210 configurations; full results in Table 4, Appendix B).

FWL invariance. Under linear OLS, for four topologies where A enters as a covariate (confounding, mediation, collider, M-bias), FWL implies that any linear transformation of the dummy matrix $\mathbf{D}^*$ (C^{-1} premultiplication, MLA, SIMEX) preserves the column space relative to T, leaving the treatment coefficient unchanged (Appendix B.2). Naive plug-in and MLA achieve exactly 0.0% bias reduction. This invariance does not extend to nonlinear models (Appendix B.6).

Method ranking. PPI++ achieves the highest grand-mean RMSE reduction (47.0%) but requires gold labels and is harmful for M-bias (−88%). MC-SIMEX provides gold-free correction (best for IV at 59.4%) but harms front-door (−62.4%). For protective topologies, correction re-exposes the original bias pathway. No single method dominates: correction effectiveness is itself topology-dependent (Appendix B.3).

5 Conclusion

We characterized how multi-class annotation errors interact with causal DAG topology across seven canonical topologies. The classical binary guarantee breaks at $K \geq 3$: ex-

posure and IV topologies produce sign reversal, while five topologies maintain a structural guarantee. The ASV framework reveals a vulnerability spectrum from IV (maximally fragile) to M-bias (immune), and matrix-based corrections are provably ineffective for four topologies due to FWL invariance. Empirical validation with 23 real LLM configurations confirms: the primary treatment effect never reverses, but 61% of exposure configurations reverse a secondary coefficient. Differential misclassification and correlated multi-LLM errors are natural extensions. Code for ASV computation and all experiments will be publicly released.

Limitations

Our framework assumes non-differential misclassification ($P(A^*|A, Y, \mathbf{Z}) = P(A^*|A)$), which holds when the LLM does not observe Y but may be violated when it accesses Y-correlated information—e.g., outcome-conditional prompting, few-shot demonstrations encoding outcome–treatment associations, or retrieval-augmented annotation carrying outcome signal (Appendix B.5). The framework assumes a correctly specified causal DAG; when the topology is uncertain, practitioners should compute ASV under candidate DAGs and report the most conservative. Closed-form expressions are derived for K=3 and verified at K=5, 7 under linear structural equations; extensions to arbitrary K and nonlinear models (Appendix B.6) remain open. We address misclassification of A by a single annotator; outcome misclassification and correlated multi-annotator settings (Kim et al., 2025) introduce additional bias channels not captured here.

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A Proofs of Proposition 1 and Theorem 1

We provide algebraic proofs of sign preservation for the confounding and collider topologies (Proposition 1(a)–(b)), an explicit counterexample for exposure (Proposition 1(c)), numerical verification for mediation, M-bias, and front-door (Numerical Verification 1), and the monotonicity proof for confounding (Theorem 1). Throughout, $A \in \{0, 1, \ldots, K-1\}$ with prior $p_k = P(A{=}k)$, and C is a $K \times K$ column-stochastic confusion matrix with $C_{jk} = P(A^*{=}j \mid A{=}k)$. We use category 0 as the reference and let $D_j = \mathbf{1}(A{=}j)$, $D_j^* = \mathbf{1}(A^*{=}j)$ for $j = 1, \ldots, K-1$.

A.1 Notation and Key Identities

Under non-differential misclassification, the misclassified moments factor linearly through C. Define the population vectors:

$$q_j = P(A^*{=}j) = \textstyle\sum_k C_{jk}\, p_k, \tag{4}$$

$$\mathrm{ct}_j = \mathbb{E}[T \cdot D_j^*] = \textstyle\sum_k C_{jk}\, \mathbb{E}[T \cdot D_k], \tag{5}$$

$$\mathrm{cy}_j = \mathbb{E}[Y \cdot D_j^*] = \textstyle\sum_k C_{jk}\, \mathbb{E}[Y \cdot D_k]. \tag{6}$$

Write $q = Cp$, $\mathrm{ct} = C\,\mathbb{E}[T \cdot D]$, and $\mathrm{cy} = C\,\mathbb{E}[Y \cdot D]$, where $D = (D_0, \ldots, D_{K-1})^\top$.

A.2 Frisch–Waugh–Lovell Decomposition

For topologies where T is directly observed (confounding, mediation, collider, M-bias, front-door), the estimator regresses Y on $Z^* = (1, T, D_1^*, \ldots, D_{K-1}^*)$. By the Frisch–Waugh–Lovell (FWL) theorem, the coefficient on T satisfies:

$$\operatorname{plim} \hat{\tau}(C) = \frac{\mathbb{E}[\tilde{T}\,\tilde{Y}]}{\mathbb{E}[\tilde{T}^2]}, \tag{7}$$

where $\tilde{T} = T - \mathbb{E}[T \mid A^*]$ and $\tilde{Y} = Y - \mathbb{E}[Y \mid A^*]$ are residuals from projecting onto $(1, D_1^*, \ldots, D_{K-1}^*)$.

Lemma 1 (Bias decomposition)**.** *Suppose $Y = \tau T + f(A) + \varepsilon_Y$ with $\varepsilon_Y \perp (T, A, A^*)$. Then*

$$\operatorname{plim} \hat{\tau}(C) = \tau + \frac{R(C)}{D(C)}, \tag{8}$$

where

$$R(C) = \mathbb{E}\big[\mathrm{Cov}(T,\, f(A) \mid A^*)\big], \tag{9}$$

$$D(C) = \mathbb{E}\big[\mathrm{Var}(T \mid A^*)\big] > 0. \tag{10}$$

Proof. Expand $\tilde{Y}$ using $Y = \tau T + f(A) + \varepsilon_Y$:

$$\tilde{Y} = \tau\,\tilde{T} + \big(f(A) - \mathbb{E}[f(A) \mid A^*]\big) + \varepsilon_Y.$$

Since $\tilde{T}$ is measurable w.r.t. (T, A^*) and $\varepsilon_Y \perp (T, A, A^*)$:

$$\mathbb{E}[\tilde{T}\,\tilde{Y}] = \tau\,\mathbb{E}[\tilde{T}^2] + \mathbb{E}\big[\tilde{T}\,\big(f(A) - \mathbb{E}[f(A) \mid A^*]\big)\big].$$

The numerator residual is $R(C) = \mathbb{E}[\tilde{T}(f(A) - \mathbb{E}[f(A) \mid A^*])]$, and the denominator is $D(C) = \mathbb{E}[\tilde{T}^2]$.

By the law of total covariance:

$$R(C) = \sum_{j=0}^{K-1} q_j\, \mathrm{Cov}(T,\, f(A) \mid A^*{=}j) = \mathbb{E}[\mathrm{Cov}(T, f(A) \mid A^*)].$$

For $D(C) > 0$: decompose $T = g(A) + \varepsilon_T$ where $g(A) = \mathbb{E}[T \mid A]$ (for confounding) or simply note that T has exogenous variation independent of A^* (for mediation). In both cases, $\tilde{T} = \big(g(A) - \mathbb{E}[g(A) \mid A^*]\big) + \varepsilon_T'$, where ε_T' is the component of T independent of (A, A^*). Hence $D(C) \geq \mathrm{Var}(\varepsilon_T') > 0$. □

A.3 Confounding Topology (A = Confounder)

Data-generating process. $A \to T$, $A \to Y$, $T \to Y$. Using dummy coding with category 0 as reference:

$$T = \sum_{j=1}^{K-1} \alpha_j D_j + \varepsilon_T, \quad \varepsilon_T \sim \mathcal{N}(0, \sigma_T^2),\ \varepsilon_T \perp A,$$

$$Y = \tau\, T + \sum_{j=1}^{K-1} \beta_j^{(A)} D_j + \varepsilon_Y, \quad \varepsilon_Y \sim \mathcal{N}(0,1),\ \varepsilon_Y \perp (T, A).$$

Define $g(k) = \sum_j \alpha_j \mathbf{1}(k{=}j)$ and $f(k) = \sum_j \beta_j^{(A)} \mathbf{1}(k{=}j)$, so $g(0) = f(0) = 0$.

Closed-form plim. Using the FWL decomposition, the coefficient on T is:

$$\operatorname{plim} \hat\tau(C) = \frac{\mu_{TY} - \sum_{j=0}^{K-1} \frac{\mathrm{ct}_j\, \mathrm{cy}_j}{q_j}}{s_T - \sum_{j=0}^{K-1} \frac{\mathrm{ct}_j^2}{q_j}}, \tag{11}$$

where $s_T = \mathbb{E}[T^2]$, $\mu_{TY} = \mathbb{E}[TY]$, and ct_j, cy_j, q_j are defined in (4)–(6).

Derivation. By FWL, $\operatorname{plim} \hat\tau = \mathbb{E}[\tilde T Y]/\mathbb{E}[\tilde T^2]$ where $\tilde T = T - \mathbb{E}[T \mid A^*]$. Since $\mathbb{E}[T \mid A^*{=}j] = \mathrm{ct}_j/q_j$:

$$\mathbb{E}[\tilde T^2] = \mathbb{E}[T^2] - \sum_j q_j \big(\mathrm{ct}_j/q_j\big)^2 = s_T - \sum_j \mathrm{ct}_j^2/q_j,$$

$$\mathbb{E}[\tilde T\, Y] = \mathbb{E}[TY] - \sum_j \big(\mathrm{ct}_j/q_j\big)\, \mathbb{E}[Y \cdot D_j^*] = \mu_{TY} - \sum_j \mathrm{ct}_j\, \mathrm{cy}_j/q_j.$$

□

Verification at $C = I$. When $C = I$: $q_j = p_j$, $\mathrm{ct}_j = \mathbb{E}[TD_j] = \alpha_j p_j$, $\mathrm{cy}_j = \mathbb{E}[YD_j] = p_j(\tau\alpha_j + \beta_j^{(A)})$. Direct substitution yields numerator $= \tau\sigma_T^2$ and denominator $= \sigma_T^2$, giving $\operatorname{plim} \hat\tau = \tau$.

Bias structure. By Lemma 1:

$$\operatorname{plim} \hat\tau(C) = \tau + B(C), \quad B(C) = \frac{\mathbb{E}[\mathrm{Cov}(g(A),\, f(A) \mid A^*)]}{D(C)}. \tag{12}$$

By the law of total covariance:

$$\mathbb{E}[\mathrm{Cov}(g, f \mid A^*)] = \mathrm{Cov}(g(A), f(A)) - \mathrm{Cov}\big(\mathbb{E}[g(A) \mid A^*],\, \mathbb{E}[f(A) \mid A^*]\big). \tag{13}$$

At $C = I$: $\mathbb{E}[g \mid A^*] = g(A)$, so the between-group covariance equals $\mathrm{Cov}(g, f)$ and $B(I) = 0$.

At $C = \frac{1}{K}\mathbf{1}\mathbf{1}^\top$ (uniform, $A^* \perp A$): $\mathbb{E}[g \mid A^*]$ is constant, so the between-group covariance is zero and

$$B_{\mathrm{OV}} \;\equiv\; B\big(\tfrac{1}{K}\mathbf{1}\mathbf{1}^\top\big) = \frac{\mathrm{Cov}(g(A),\, f(A))}{\mathrm{Var}(T)}, \tag{14}$$

which equals the classical omitted-variable bias (no control for A).

Sign preservation condition. Since $D(C) > 0$, $\operatorname{sign}(\operatorname{plim} \hat\tau) = \operatorname{sign}(\tau \cdot D(C) + R(C))$. For sign preservation, it suffices that $R(C)$ does not overwhelm $\tau \cdot D(C)$.

Proposition 2 (Confounding sign preservation — proportional case)**.** *If $f(A) = \lambda\, g(A)$ for some constant λ (i.e., the confounder effects on T and Y are proportional), then*

$$B(C) = \big(1 - \eta(C)\big)\, B_{\mathrm{OV}}, \tag{15}$$

where

$$\eta(C) = \frac{\mathrm{Var}(\mathbb{E}[g(A) \mid A^*])\, \sigma_T^2}{\mathrm{Var}(g(A))\big(\mathrm{Var}(g(A)) - \mathrm{Var}(\mathbb{E}[g(A) \mid A^*]) + \sigma_T^2\big)} \in [0, 1].$$

Consequently, $B(C)$ lies between 0 and B_{OV}, and $\operatorname{sign}(\operatorname{plim} \hat\tau) = \operatorname{sign}(\tau)$ whenever $\operatorname{sign}(\tau) = \operatorname{sign}(\tau + B_{\mathrm{OV}})$.

Proof. When $f = \lambda g$: the between-group covariance in (13) becomes $\lambda\, \mathrm{Var}(\mathbb{E}[g \mid A^*])$, and $\mathrm{Cov}(g, f) = \lambda\, \mathrm{Var}(g)$. So $R(C) = \lambda\big(\mathrm{Var}(g) - \mathrm{Var}(\mathbb{E}[g \mid A^*])\big) = \lambda V_W$, where $V_W = \mathrm{Var}(g) - V_B$ with $V_B = \mathrm{Var}(\mathbb{E}[g \mid A^*])$. Similarly, $D(C) = V_W + \sigma_T^2$. Hence

$$B(C) = \frac{\lambda\, V_W}{V_W + \sigma_T^2}, \qquad B_{\mathrm{OV}} = \frac{\lambda\, V}{V + \sigma_T^2},$$

where $V = \mathrm{Var}(g) = V_B + V_W$. The ratio

$$\frac{B(C)}{B_{\mathrm{OV}}} = \frac{V_W(V + \sigma_T^2)}{V(V_W + \sigma_T^2)} = 1 - \eta(C)$$

is a decreasing function of $V_B \in [0, V]$. At $V_B = 0$ (no classification, C uniform): $\eta = 0$, $B = B_{\mathrm{OV}}$. At $V_B = V$ (perfect classification, $C = I$): $\eta = 1$, $B = 0$. Since $0 \le V_B \le V$, we have $\eta \in [0, 1]$ and $B(C) \in [0, B_{\mathrm{OV}}]$ (with the sign of B_{OV}). □

Proposition 3 (Confounding sign preservation — general case)**.** *For general g and f (not*

necessarily proportional), define the worst-case bias bound

$$\bar{B} = \frac{\sqrt{\mathrm{Var}(g(A))\ \mathrm{Var}(f(A))}}{\sigma_T^2}. \tag{16}$$

If $|\tau| > \bar{B}$*, then* $\mathrm{sign}(\mathrm{plim}\,\hat{\tau}(C)) = \mathrm{sign}(\tau)$ *for* all *column-stochastic* C*.*

Proof. We need $\mathrm{sign}(\tau \cdot D(C) + R(C)) = \mathrm{sign}(\tau)$, which holds if $|R(C)| < |\tau| \cdot D(C)$. Since $D(C) \geq \sigma_T^2$ (shown in Lemma 1), it suffices to show $|R(C)| < |\tau|\,\sigma_T^2$.

Step 1: Bound $|R(C)|$. $R(C) = \mathbb{E}[\mathrm{Cov}(g(A), f(A) \mid A^*)]$. By the Cauchy–Schwarz inequality applied to each within-group covariance:

$$|\,\mathrm{Cov}(g, f \mid A^*{=}j)| \leq \sqrt{\mathrm{Var}(g \mid A^*{=}j)\ \mathrm{Var}(f \mid A^*{=}j)}.$$

Taking expectations and applying Cauchy–Schwarz again:

$$\begin{aligned} |R(C)| &\leq \mathbb{E}\big[\sqrt{\mathrm{Var}(g \mid A^*)\ \mathrm{Var}(f \mid A^*)}\big] \\ &\leq \sqrt{\mathbb{E}[\mathrm{Var}(g \mid A^*)]\,\mathbb{E}[\mathrm{Var}(f \mid A^*)]} \\ &\leq \sqrt{\mathrm{Var}(g(A))\ \mathrm{Var}(f(A))}, \end{aligned}$$

where the last step uses $\mathbb{E}[\mathrm{Var}(X \mid Y)] \leq \mathrm{Var}(X)$ (law of total variance).

Step 2: Conclude. The bias satisfies

$$|B(C)| = \frac{|R(C)|}{D(C)} \leq \frac{\sqrt{\mathrm{Var}(g)\ \mathrm{Var}(f)}}{\sigma_T^2} = \bar{B}.$$

When $|\tau| > \bar{B}$, the bias cannot cancel τ, so $\mathrm{sign}(\mathrm{plim}\,\hat{\tau}) = \mathrm{sign}(\tau)$ for all C. □

Relationship to the omitted-variable condition. A weaker (necessary) condition for sign preservation is $\mathrm{sign}(\tau) = \mathrm{sign}(\tau_{\mathrm{OV}})$ where $\tau_{\mathrm{OV}} = \tau + \mathrm{Cov}(g, f)/\,\mathrm{Var}(T)$. If this fails, then $C = \frac{1}{K}\mathbf{1}\mathbf{1}^\top$ (uniform noise) already flips the sign. The rigorous sufficient condition $|\tau| > \bar{B}$ implies the weaker condition, since $|B_{\mathrm{OV}}| = |\,\mathrm{Cov}(g, f)|/\,\mathrm{Var}(T) \leq \bar{B}$ by Cauchy–Schwarz and $\mathrm{Var}(T) \geq \sigma_T^2$.

Verification for paper parameters (K=3). With τ=2.0, $\boldsymbol{\alpha}$=(1.5, −1.0), $\boldsymbol{\beta}^{(A)}$=(1.0, −0.5), σ_T=1.0, $\mathbf{p}$=(0.40, 0.35, 0.25):

$$\mathrm{Var}(g) = \sum_k p_k \alpha_k^2 - \big(\textstyle\sum_k p_k \alpha_k\big)^2 = 0.962,$$

$$\mathrm{Var}(f) = \sum_k p_k (\beta_k^{(A)})^2 - \big(\textstyle\sum_k p_k \beta_k^{(A)}\big)^2 = 0.362,$$

$$\bar{B} = \sqrt{0.962 \times 0.362}\ /\ 1.0^2 = 0.590.$$

Since $|\tau|$=2.0 > $\bar{B}$=0.590, sign preservation is guaranteed for *all* column-stochastic C by Proposition 3. Additionally, $B_{\mathrm{OV}} = \mathrm{Cov}(g, f)/\,\mathrm{Var}(T) = 0.588/1.962 = +0.300$, giving τ_{OV}=2.300 > 0, confirming that both the necessary and sufficient conditions are satisfied.

Extension to $K > 3$. For K>3, treatment effect coefficients are drawn from a fixed pool with alternating signs and decreasing magnitudes: $\boldsymbol{\beta} = (0, 1.0, -0.5, 0.7, -0.3, 0.5, -0.2, 0.4, -0.15, 0.3, \ldots)$, truncated to K entries. The category distribution uses $p_k \propto 1/(k{+}1)$, normalized to sum to 1. Other topology-specific parameters (confounding strengths, mediator effects, etc.) follow analogous alternating-sign pools. For K=10, $\boldsymbol{\beta} = (0, 1.0, -0.5, 0.7, -0.3, 0.5, -0.2, 0.4, -0.15, 0.3)$.

A.4 Proof of Theorem 1 (Confounding Monotonicity)

Under the interpolation $C(\delta) = (1{-}\delta)I + \delta C_0$, we prove $|B(C(\delta))|$ is non-decreasing in δ for proportional effects ($f = \lambda g$) and provide numerical verification for the general case. The proof proceeds by direct algebraic analysis of $V_B(\delta) = \mathrm{Var}(\mathbb{E}[g(A) \mid A^*_\delta])$ as a rational function of δ, establishing convexity and bounding the derivative at both endpoints.

Proportional case ($f = \lambda g$). From Proposition 2, $B(C) = \lambda V_W/(V_W + \sigma_T^2)$ where $V_W = \mathrm{Var}(g) - V_B$ and $V_B(\delta) = \mathrm{Var}(\mathbb{E}[g(A) \mid A^*_\delta])$.

Step 1: Convexity of $V_B(\delta)$. Define $N_j(\delta) = \sum_k g_k\, C(\delta)_{jk}\, p_k$ and $q_j(\delta) = \sum_k C(\delta)_{jk}\, p_k$. Both are affine in δ, with $q_j(\delta) > 0$. Then $V_B(\delta) = \sum_j N_j(\delta)^2/q_j(\delta) - \bar{g}^2$. Each term N_j^2/q_j is the composition of the perspective function $(x, t) \mapsto x^2/t$ (convex for $t > 0$) with the affine map $\delta \mapsto (N_j(\delta), q_j(\delta))$, hence convex in δ. A sum of convex functions is convex, so $V_B(\delta)$ is convex on $[0, \delta_{\max}]$.

Step 2: Derivative formula. Let $\mu_j(\delta) = N_j(\delta)/q_j(\delta) = \mathbb{E}[g(A) \mid A^*{=}j]$. Then

$$\frac{dV_B}{d\delta} = -\sum_{j,k} \big[(C_0)_{jk} - \mathbf{1}_{j=k}\big]\, p_k\, (g_k - \mu_j(\delta))^2. \tag{17}$$

Derivation. $\frac{d}{d\delta}\frac{N_j^2}{q_j} = 2\mu_j \dot{N}_j - \mu_j^2 \dot{q}_j$ where $\dot{N}_j = m_j^0 - g_j p_j$ and $\dot{q}_j = q_j^0 - p_j$ (with

$m_j^0 = \sum_k g_k (C_0)_{jk} p_k$, $q_j^0 = \sum_k (C_0)_{jk} p_k$). Substituting and using column-stochasticity ($\sum_j (C_0)_{jk} = \sum_j \mathbf{1}_{j=k} = 1$) to cancel the g_k^2 terms yields (17).

Step 3: $V_B'(0) \le 0$ (universal, no conditions on C_0). At δ=0: $\mu_j(0) = g_j$ (since $C(0) = I$), so $(g_k - \mu_j(0))^2 = (g_k - g_j)^2$. The diagonal terms ($j$=$k$) vanish. Hence

$$V_B'(0) = -\sum_{j \neq k} (C_0)_{jk}\, p_k\, (g_j - g_k)^2 \;\le\; 0,$$

since every term is non-negative.

Step 4: $V_B'(1) \le 0$ for K=2. For K=2 with $C_0 = \left[\begin{smallmatrix} a & 1-b \\ 1-a & b \end{smallmatrix}\right]$ and $a, b \ge \frac{1}{2}$, set g_0=0, g_1=1 without loss of generality (the general case scales by $(\Delta g)^2$). At δ=1: q_0=ap_0+$(1-b)p_1$, q_1=$(1-a)p_0$+bp_1, μ_0=$(1-b)p_1/q_0$, μ_1=bp_1/q_1. Expanding (17) at δ=1 and factoring:

$$\begin{aligned} V_B'(1) &= (1-a)\, p_0(\mu_0^2 - \mu_1^2) + (1-b)\, p_1\big[(1-\mu_1)^2 - (1-\mu_0)^2\big] \\ &= (\mu_0 - \mu_1)\, \big[(1-a)\, p_0(\mu_0 + \mu_1) + (1-b)\, p_1(2 - \mu_0 - \mu_1)\big]. \end{aligned}$$

The bracket is non-negative (since $0 \le \mu_j \le 1$ and all coefficients are non-negative). Direct calculation gives $\mu_0 - \mu_1 = p_0\, p_1(1-a-b)/(q_0\, q_1)$ (expand $(1-b)\, q_1 - b\, q_0 = p_0(1-a-b)$). Under $a, b \ge \frac{1}{2}$: $a+b \ge 1$, so $\mu_0 - \mu_1 \le 0$ and $V_B'(1) \le 0$.

Step 5: Monotonicity from convexity. Since V_B is convex (Step 1), its derivative V_B' is non-decreasing. When $V_B'(1) \le 0$, we have $V_B'(\delta) \le V_B'(1) \le 0$ for all $\delta \in [0, 1]$, so V_B is non-increasing.

Step 6: $|B(\delta)|$ is non-decreasing. $V_W(\delta) = \mathrm{Var}(g) - V_B(\delta)$ is non-decreasing (Step 5). The function $h(x) = |\lambda|\, x/(x + \sigma_T^2)$ is strictly increasing for $x \ge 0$. Hence $|B(\delta)| = h(V_W(\delta))$ is non-decreasing. □

Condition on C_0. The proof requires $V_B'(1) \le 0$, which is guaranteed for K=2 when $(C_0)_{jj} \ge \frac{1}{2}$ (Step 4). For $K \ge 3$, $V_B'(1) \le 0$ holds for all g and p if and only if a certain matrix $M - W$ (depending on C_0 and p) is negative semi-definite; simple diagonal thresholds are sufficient in practice but not universal. Numerical verification across 20,000 random (C_0, p, g) triples per $K \in \{3, 5, 7\}$ with $(C_0)_{jj} \ge 1/K$ finds worst-case V_B increases $<$ 0.005, confirming ε-approximate monotonicity. A K=2 counterexample (C_0 the permutation matrix, $(C_0)_{jj}$=0) shows the diagonal condition is necessary: $V_B(\delta) = (\delta - \frac{1}{2})^2$ increases on $(\frac{1}{2}, 1]$.

General case (non-proportional effects). $|B(C(\delta))| \le \bar{B}(\delta)$ where $\bar{B}(\delta) = \sqrt{\mathbb{E}[\mathrm{Var}(g \mid A_\delta^*)]\, \mathbb{E}[\mathrm{Var}(f \mid A_\delta^*)]}\, /\, D(\delta)$ (Proposition 3). By the same convexity argument, each $\mathbb{E}[\mathrm{Var}(\cdot \mid A_\delta^*)]$ is non-decreasing in δ, and $D(\delta) = \mathbb{E}[\mathrm{Var}(T \mid A_\delta^*)]$ is non-decreasing. Extended verification (20,000 random C_0 matrices, $K \in \{3, 5, 7\}$) reveals worst-case step deviations of 0.011 (confounding), confirming approximate tightness; a strict monotonicity proof for the general case remains open.

A.5 Mediation Topology (A = Mediator)

Data-generating process. $T \to A \to Y$, $T \to Y$.

$$\begin{aligned} T &\sim \mathrm{Bernoulli}(0.5), \\ A \mid T &\sim \mathrm{Categorical}\big(\mathrm{softmax}(\boldsymbol{\gamma} + \boldsymbol{\delta}\, T)\big), \\ Y &= \tau_{\mathrm{dir}}\, T + f(A) + \varepsilon_Y. \end{aligned}$$

Bias decomposition. By Lemma 1, the direct-effect estimator satisfies:

$$\mathrm{plim}\, \hat{\tau}_{\mathrm{dir}}(C) = \tau_{\mathrm{dir}} + \frac{\mathbb{E}[\mathrm{Cov}(T,\, f(A) \mid A^*)]}{D(C)}. \tag{18}$$

The key structural difference from confounding: T *causes* A, so the residual covariance $\mathrm{Cov}(T, f(A) \mid A^*)$ captures the indirect effect of T on Y through A that is not absorbed by conditioning on A^*.

Endpoint analysis. At $C = I$ (A^*=A): conditioning on A blocks the mediated path $T \to A \to Y$, so $\mathrm{Cov}(T, f(A) \mid A) = 0$ and $\mathrm{plim}\, \hat{\tau} = \tau_{\mathrm{dir}}$.

At $C = \frac{1}{K}\mathbf{1}\mathbf{1}^\top$ ($A^* \perp A$): no information about A is used, and

$$\begin{aligned} \mathbb{E}[\mathrm{Cov}(T, f(A) \mid A^*)] &= \mathrm{Cov}(T, f(A)), \\ D(C) &= \mathrm{Var}(T) = 0.25. \end{aligned}$$

The indirect effect is $\tau_{\mathrm{ind}} = \mathbb{E}[f(A) \mid T$=$1] - \mathbb{E}[f(A) \mid T$=$0]$, and

$$\mathrm{Cov}(T, f(A)) = 0.25\, \tau_{\mathrm{ind}}.$$

Hence at C uniform: $\mathrm{plim}\, \hat{\tau} = \tau_{\mathrm{dir}} + \tau_{\mathrm{ind}} = \tau_{\mathrm{total}}$.

Numerical Verification (Mediation sign preservation). $\operatorname{sign}(\operatorname{plim}\hat{\tau}_{\text{dir}}) = \operatorname{sign}(\tau_{\text{dir}})$ whenever $\operatorname{sign}(\tau_{\text{dir}}) = \operatorname{sign}(\tau_{\text{total}})$, where $\tau_{\text{total}} = \tau_{\text{dir}} + \tau_{\text{ind}}$. This result is verified numerically for 100,000 independently sampled column-stochastic matrices C; a closed-form proof for general K remains open.

Proof. Let $R(C) = \mathbb{E}[\operatorname{Cov}(T, f(A) \mid A^*)]$ and $D(C) = \mathbb{E}[\operatorname{Var}(T \mid A^*)] > 0$.

Endpoints. At $C{=}I$: $f(A)$ is constant given $A{=}A^*$, so $R(I) = 0$ and $B(I) = 0$. At $C = \frac{1}{K}\mathbf{1}\mathbf{1}^\top$: $R = \operatorname{Cov}(T, f(A)) = 0.25\,\tau_{\text{ind}}$ and $D = \operatorname{Var}(T) = 0.25$, giving $B = \tau_{\text{ind}}$.

Within-group structure for binary T. Since $T \in \{0, 1\}$, the within-A^*-group bias admits a revealing decomposition. Let $p_j = P(T{=}1 \mid A^*{=}j)$. Then $\operatorname{Var}(T \mid A^*{=}j) = p_j(1{-}p_j)$ and

$$\operatorname{Cov}(T, f(A) \mid A^*{=}j) = p_j(1{-}p_j)\big[\mathbb{E}[f(A) \mid T{=}1, A^*{=}j] - \mathbb{E}[f(A) \mid T{=}0, A^*{=}j]\big]$$

Define $\delta_j = \mathbb{E}[f(A) \mid T{=}1, A^*{=}j] - \mathbb{E}[f(A) \mid T{=}0, A^*{=}j]$ as the *conditional indirect effect* in the $A^*{=}j$ subpopulation. Then

$$B(C) = \frac{R(C)}{D(C)} = \frac{\sum_j q_j\, p_j(1{-}p_j)\,\delta_j}{\sum_j q_j\, p_j(1{-}p_j)} = \sum_j w_j\,\delta_j,$$

a weighted average of the conditional indirect effects, with weights $w_j = q_j p_j(1{-}p_j) / \sum_\ell q_\ell p_\ell(1{-}p_\ell) \geq 0$ summing to 1.

At $C{=}I$: each $A^*{=}j$ group has $A{=}j$ a.s., so $\delta_j = f(j) - f(j) = 0$ and $B = 0$. At C uniform: $\delta_j = \tau_{\text{ind}}$ for all j and $B = \tau_{\text{ind}}$.

Sign preservation. When $\operatorname{Cov}(g(A), f(A)) > 0$ (the indirect effect reinforces the direct effect, i.e., τ_{dir} and τ_{ind} have the same sign), the bias $B(C) \geq 0$ cannot oppose τ_{dir}, so sign is preserved trivially.

When τ_{dir} and τ_{ind} have opposite signs but $|\tau_{\text{ind}}| < |\tau_{\text{dir}}|$ (i.e., $\operatorname{sign}(\tau_{\text{dir}}) = \operatorname{sign}(\tau_{\text{total}})$), sign reversal requires some $|\delta_j| > |\tau_{\text{dir}}|$ with the opposing sign. Since each δ_j is a difference of weighted averages of f values (with weights $P(A{=}k \mid T{=}t, A^*{=}j)$), and $B(C)$ is a convex combination of δ_j, the bound $|B(C)| \leq \max_j |\delta_j|$ holds. We verify numerically that $\max_j |\delta_j| \leq |\tau_{\text{ind}}|$ for all 100,000 random C matrices in our experiments, confirming sign preservation. □

Verification for paper parameters ($K{=}3$). With $\tau_{\text{dir}}{=}0.5$, $\boldsymbol{\gamma}{=}(0, 0, 0)$, $\boldsymbol{\delta}{=}(0, 0.8, -0.6)$, $\boldsymbol{\beta}{=}(0, 1.0, -0.5)$:

$\tau_{\text{ind}} = \mathbb{E}[f(A) \mid T{=}1] - \mathbb{E}[f(A) \mid T{=}0] = 0.351$ (computed from the softmax class probabilities). $\tau_{\text{total}} = 0.5 + 0.351 = 0.851 > 0$. Since $\tau_{\text{dir}}{>}0$ and $\tau_{\text{total}}{>}0$, sign preservation holds for all C.

Mediation monotonicity (proportional effects). Under the interpolation $C(\delta) = (1{-}\delta)I + \delta C_0$, we prove $|B(C(\delta))|$ is non-decreasing in δ when the mediator's effect on Y is proportional to its informativeness about T: $f(A) = \lambda\, g(A)$ for some scalar λ, where $g(k) = \mathbb{E}[T \mid A{=}k]$.

Step 1: Decompose T via the mediator. Define $g(A) = \mathbb{E}[T \mid A]$ and $\varepsilon_T = T - g(A)$. By construction, $\mathbb{E}[\varepsilon_T \mid A] = 0$. By non-differential misclassification ($T \perp A^* \mid A$): $\mathbb{E}[\varepsilon_T \mid A^*{=}j] = \sum_k P(A{=}k \mid A^*{=}j)\,\mathbb{E}[\varepsilon_T \mid A{=}k, A^*{=}j] = 0$, so $\mathbb{E}[\varepsilon_T \mid A^*{=}j] = 0$ and $\operatorname{Cov}(\varepsilon_T,\, g(A) \mid A^*{=}j) = 0$ for every j.

Step 2: Reduce the numerator. Since $f = \lambda g$:

$$R(C) = \lambda\, \mathbb{E}[\operatorname{Var}(g(A) \mid A^*)] = \lambda\, V_W.$$

Step 3: Constant additive term in denominator.

$$D(C) = \mathbb{E}[\operatorname{Var}(T \mid A^*)] = V_W + \sigma_\varepsilon^2,$$

where $\sigma_\varepsilon^2 \equiv \mathbb{E}[\operatorname{Var}(\varepsilon_T \mid A^*)]$. This is *constant* (independent of C): $\mathbb{E}[\operatorname{Var}(\varepsilon_T \mid A^*)] = \sum_k p_k \operatorname{Var}(T \mid A{=}k) = \mathbb{E}[\operatorname{Var}(T \mid A)]$, where the equality uses column-stochasticity: $\sum_j q_j\, P(A{=}k \mid A^*{=}j) = p_k$.

Step 4: Closed form and monotonicity. $B(C) = \lambda\, V_W / (V_W + \sigma_\varepsilon^2)$, structurally identical to confounding (Proposition 2) with σ_ε^2 playing the role of σ_T^2. Since $V_B(\delta)$ is convex with $V_B'(0) \leq 0$ (same argument as §A.4, Steps 1–3), $V_W(\delta) = V - V_B(\delta)$ is non-decreasing under the same conditions on C_0. Since $h(x) = |\lambda| x/(x + \sigma_\varepsilon^2)$ is increasing, $|B(C(\delta))|$ is non-decreasing.

Boundedness of the leakage coefficient. The mediation bias (3) writes the residual indirect effect as $(1 - \lambda(C))\,\tau_{\text{ind}}$, which (for $\tau_{\text{ind}} \neq 0$) *implicitly* defines

$$\lambda(C) \;:=\; 1 - \frac{B(C)}{\tau_{\text{ind}}}, \qquad B(C) \;\equiv\; \operatorname{plim}\hat{\tau}_{\text{dir}}(A^*) - \tau_{\text{dir}}, \quad \tau_{\text{ind}} \neq 0 \tag{19}$$

Substituting back into (3) gives the convex-combination form $\operatorname{plim}\hat{\tau}_{\text{dir}}(C) = \lambda(C)\,\tau_{\text{dir}} + (1-\lambda(C))\,\tau_{\text{total}}$, so $\lambda(C)$ is the weight placing the biased plim between τ_{dir} (perfect classification) and τ_{total} (no information).

Proposition 4 (Mediation leakage boundedness)**.** *Let $\lambda(C)$ be defined by (19) with $\tau_{\text{ind}} \neq 0$, and let C be column-stochastic.*

(a) $\lambda(C) \leq 1$ iff $\operatorname{sign}(B(C)) = \operatorname{sign}(\tau_{\text{ind}})$ (the bias points in the direction of the indirect effect; no overcorrection).

(b) $\lambda(C) \geq 0$ iff additionally $|B(C)| \leq |\tau_{\text{ind}}|$ (the bias does not exceed the indirect effect in magnitude).

Equivalently, $\lambda(C) \in [0,1]$ iff $\operatorname{plim}\hat{\tau}_{\text{dir}}(C) \in [\min(\tau_{\text{dir}}, \tau_{\text{total}}), \max(\tau_{\text{dir}}, \tau_{\text{total}})]$.

Under proportional effects *($f(A) = \lambda_0\, g(A)$ with $g(k) = \mathbb{E}[T \mid A{=}k]$), both (a) and (b) hold for every C, so $\lambda(C) \in [0,1]$ unconditionally. For* general (non-proportional) *f, both (a) and (b) can fail because $\operatorname{Cov}(\mathbb{E}[g \mid A^*], \mathbb{E}[f \mid A^*])$ may differ in sign or magnitude from $\operatorname{Cov}(g, f)$ (a covariance-level analogue of Simpson's paradox under A^* aggregation); the numerical verification below quantifies the empirical rate. The endpoints $\lambda(I) = 1$ and $\lambda(C) \to 0$ in the rank-one (uninformative) limit hold in all regimes.*

Sign preservation. *When $\lambda(C) \in [0,1]$, the convex-combination form gives $\operatorname{plim}\hat{\tau}_{\text{dir}}(C) \in [\min(\tau_{\text{dir}}, \tau_{\text{total}}), \max(\tau_{\text{dir}}, \tau_{\text{total}})]$. If additionally $\operatorname{sign}(\tau_{\text{dir}}) = \operatorname{sign}(\tau_{\text{total}})$, both endpoints share the same sign, so $\operatorname{sign}(\operatorname{plim}\hat{\tau}_{\text{dir}}) = \operatorname{sign}(\tau_{\text{dir}})$ for every C. Under proportional effects, $\lambda(C) \in [0,1]$ is automatic (above), so sign preservation of $\hat{\tau}_{\text{dir}}$ reduces to the single condition $\operatorname{sign}(\tau_{\text{dir}}) = \operatorname{sign}(\tau_{\text{total}})$.*

Proof. (a),(b). With $\tau_{\text{ind}} \neq 0$, $\lambda(C) \leq 1 \iff B(C)/\tau_{\text{ind}} \geq 0 \iff \operatorname{sign}(B(C)) = \operatorname{sign}(\tau_{\text{ind}})$. Given (a), $\lambda(C) \geq 0 \iff B(C)/\tau_{\text{ind}} \leq 1 \iff |B(C)| \leq |\tau_{\text{ind}}|$.

Proportional case. Step 4 of the mediation derivation gives $B(C) = \lambda_0\, V_W(C)/(V_W(C) + \sigma_\varepsilon^2)$; the rank-one limit $V_W \to V$ yields $\tau_{\text{ind}} = \lambda_0\, V/(V + \sigma_\varepsilon^2)$. Both quantities carry $\operatorname{sign}(\lambda_0)$, establishing (a). The ratio

$$\frac{B(C)}{\tau_{\text{ind}}} = \frac{V_W(C)\,(V+\sigma_\varepsilon^2)}{V\,(V_W(C)+\sigma_\varepsilon^2)}$$

is monotone increasing in $V_W(C) \in [0, V]$ with range $[0,1]$ (its derivative in V_W is $\sigma_\varepsilon^2(V + \sigma_\varepsilon^2)/[V(V_W + \sigma_\varepsilon^2)^2] > 0$), establishing (b). Hence $\lambda(C) \in [0,1]$ unconditionally.

General case. Substituting $\lambda(C)$ from (19) into (3) yields $\operatorname{plim}\hat{\tau}_{\text{dir}}(C) = \lambda(C)\,\tau_{\text{dir}} + (1-\lambda(C))\,\tau_{\text{total}}$, so $\lambda(C) \in [0,1]$ iff $\operatorname{plim}\hat{\tau}_{\text{dir}}(C)$ lies in the closed interval with endpoints τ_{dir} and τ_{total}.

Endpoints. At $C{=}I$, $A^*{=}A$ a.s., $\mathbb{E}[T \mid A^*] = g(A)$, and FWL gives $\operatorname{plim}\hat{\tau}_{\text{dir}} = \tau_{\text{dir}}$, hence $B(I) = 0$ and $\lambda(I) = 1$. At any rank-one C (rows identical), $A^* \perp A$ so $\mathbb{E}[T \mid A^*]$ is constant, the partial regression on A^* has no effect, and $\operatorname{plim}\hat{\tau}_{\text{dir}} = \operatorname{Cov}(Y,T)/\operatorname{Var}(T) = \tau_{\text{total}}$, giving $B = \tau_{\text{ind}}$ and $\lambda = 0$. □

Relation to the between-variance ratio. A natural *diagnostic* closely related to $\lambda(C)$ is the between-variance ratio

$$\lambda_V(C) := \frac{V_B(C)}{V}, \qquad V_B(C) = \operatorname{Var}\big(\mathbb{E}[g(A) \mid A^*]\big), \quad V = \tag{20}$$

which satisfies $\lambda_V(C) \in [0,1]$ *unconditionally* by the law of total variance applied to $g(A)$. Under proportional effects, λ and λ_V agree at the endpoints and are related by

$$1-\lambda(C) = \frac{V_W(C)\,(V+\sigma_\varepsilon^2)}{V\,(V_W(C)+\sigma_\varepsilon^2)} = \big(1-\lambda_V(C)\big)\cdot\frac{V+\sigma_\varepsilon^2}{V_W(C)+\sigma_\varepsilon^2},$$

so they coincide in the limit $\sigma_\varepsilon^2 \gg V$. They generally differ for non-proportional f. We use λ throughout the main text because it parameterizes (3) directly; λ_V remains a useful summary of the channel quality of A^* as a proxy for A (independent of the structural form of f).

Numerical verification. For each $K \in \{3,5,7\}$ we draw $N{=}10{,}000$ configurations $(\pi, g, \sigma_\varepsilon^2, C)$ with $\pi \sim \text{Dirichlet}(\mathbf{1})$, $g \sim \mathcal{N}(0, I_K)$, $\sigma_\varepsilon^2 \sim \text{Uniform}(0.1, 2)$, columns of $C \sim \text{Dirichlet}(\mathbf{1})$, and evaluate $\lambda(C)$ from (19) in two regimes for f:

- **Proportional** ($f = \lambda_0\, g$, $\lambda_0 \sim \mathcal{N}(0,1) \setminus \{0\}$): $\lambda(C) \in [0,1]$ in *all* 30,000 trials (pooled observed range $[1.2{\times}10^{-5}, 0.850]$; zero out-of-bounds), confirming the closed-form bound. Sign preservation of $\hat{\tau}_{\text{dir}}$ therefore reduces to $\operatorname{sign}(\tau_{\text{dir}}) = \operatorname{sign}(\tau_{\text{total}})$.

- **Non-proportional** ($f \sim \mathcal{N}(0, I_K)$ drawn independently of g, restricted to $|\tau_{\text{ind}}| \geq 0.05$ to avoid the ill-conditioned regime where $\lambda{=}1{-}B/\tau_{\text{ind}}$ diverges): $\lambda(C)$ exceeds $[0,1]$ in $\sim$ 15–17% of trials, illustrating that (a) and (b) are not automatic outside the proportional case.

The diagnostic $\lambda_V(C)$ lies in $[0,1]$ across all 30,000 trials irrespective of regime, as guaranteed by the law of total variance. The script and full results are in `artifacts/verify_lambda_bound.py` and `artifacts/verify_lambda_bound_results.json`.

A.6 Collider Topology (A = Collider)

Data-generating process. $T \to A \leftarrow Y$, $T \to Y$. The treatment effect is unconfounded:

$$T \sim \mathcal{N}(0,1), \quad Y = \tau\, T + \varepsilon_Y, \quad \varepsilon_Y \sim \mathcal{N}(0,1),\ \varepsilon_Y \perp T,$$
$$A \mid T, Y \sim \text{Categorical}\big(\text{softmax}(\boldsymbol{\delta}_T\, T + \boldsymbol{\delta}_Y\, Y)\big).$$

The collider A is a common *effect* of T and Y. The no-control estimator (regressing Y on T alone) is consistent: $\operatorname{plim} \hat{\tau}_{\text{nc}} = \tau$. The researcher erroneously controls for A^*, inducing Berkson's bias.

Bias decomposition. Unlike confounding, ε_Y is not independent of A (since A depends on $Y = \tau T + \varepsilon_Y$), so Lemma 1 does not apply directly. Instead, we derive the bias from the FWL identity (7). Writing $\tilde{Y} = \tau\tilde{T} + (\varepsilon_Y - \mathbb{E}[\varepsilon_Y \mid A^*])$ and noting that $\mathbb{E}[\tilde{T}\,\varepsilon_Y] = \mathbb{E}[\text{Cov}(T, \varepsilon_Y \mid A^*)]$ (since $T \perp \varepsilon_Y$ unconditionally):

$$\operatorname{plim} \hat{\tau}(C) = \tau + B(C), \quad B(C) = \frac{\mathbb{E}[\text{Cov}(T,\, \varepsilon_Y \mid A^*)]}{\mathbb{E}[\text{Var}(T \mid A^*)]}. \tag{21}$$

Under non-differential misclassification ($A^* \perp (T, \varepsilon_Y) \mid A$), define $g(k) = \mathbb{E}[T \mid A{=}k]$ and $h(k) = \mathbb{E}[\varepsilon_Y \mid A{=}k]$. By the tower property, $\mathbb{E}[T \mid A^*] = \mathbb{E}[g(A) \mid A^*]$ and $\mathbb{E}[\varepsilon_Y \mid A^*] = \mathbb{E}[h(A) \mid A^*]$. Since $\text{Cov}(T, \varepsilon_Y) = 0$, the law of total covariance gives:

$$B(C) = \frac{-\,\text{Cov}\big(\mathbb{E}[g(A) \mid A^*],\, \mathbb{E}[h(A) \mid A^*]\big)}{\mathbb{E}[\text{Var}(T \mid A^*)]}.$$

Endpoint analysis. At $C = I$ ($A^*{=}A$): $B(I) = -\,\text{Cov}(g(A), h(A)) \,/\, \mathbb{E}[\text{Var}(T \mid A)] = B_{\text{Berkson}}$, the full Berkson's bias from conditioning on the collider.

At $C = \frac{1}{K}\mathbf{1}\mathbf{1}^\top$ ($A^* \perp A$): $\mathbb{E}[g(A) \mid A^*]$ and $\mathbb{E}[h(A) \mid A^*]$ are constant, so $B = 0$ and $\operatorname{plim} \hat{\tau} = \tau$.

This is the structural reverse of confounding: misclassification attenuates the collider-induced bias toward zero, rather than growing bias from zero toward B_{OV}.

Proposition 5 (Collider sign preservation — proportional case)**.** *If $h(A) = \lambda\, g(A)$ for some constant λ, then $|B(C)| \leq |B(I)|$ for all column-stochastic C, and $B(C)$ has the same sign as $B(I)$. Consequently,* $\text{sign}(\operatorname{plim} \hat{\tau}) = \text{sign}(\tau)$ *whenever* $\text{sign}(\tau) = \text{sign}(\tau + B_{\text{Berkson}})$.

Proof. When $h = \lambda g$, the numerator becomes $-\lambda\, \text{Var}(\mathbb{E}[g(A) \mid A^*])$ and the denominator is $\text{Var}(T) - \text{Var}(\mathbb{E}[g(A) \mid A^*])$. Let $V^* = \text{Var}(\mathbb{E}[g(A) \mid A^*]) \in [0, V_g]$ where $V_g = \text{Var}(g(A))$. Then

$$|B(C)| = |\lambda|\, \frac{V^*}{\text{Var}(T) - V^*}.$$

The function $x \mapsto x/(a-x)$ is increasing on $[0, a)$. With $a = \text{Var}(T) > V_g \geq V^*$: $|B(C)| \leq |\lambda|\, V_g/(\text{Var}(T) - V_g) = |B(I)|$. At $V^*{=}0$ (uniform C): $B{=}0$. At $V^*{=}V_g$ ($C{=}I$): $B{=}B(I)$. □

Proposition 6 (Collider sign preservation — general case)**.** *Define*

$$\bar{B}_{\text{col}} = \frac{\sqrt{\text{Var}(g(A))\ \text{Var}(h(A))}}{\mathbb{E}[\text{Var}(T \mid A)]}. \tag{22}$$

If $|\tau| > \bar{B}_{\text{col}}$, then $\text{sign}(\operatorname{plim} \hat{\tau}(C)) = \text{sign}(\tau)$ *for all column-stochastic C.*

Proof. Step 1: Bound the numerator. By Cauchy–Schwarz: $|\,\text{Cov}(\mathbb{E}[g \mid A^*], \mathbb{E}[h \mid A^*])| \leq \sqrt{\text{Var}(\mathbb{E}[g \mid A^*])\ \text{Var}(\mathbb{E}[h \mid A^*])}$. Since conditional expectation reduces variance: $\text{Var}(\mathbb{E}[g \mid A^*]) \leq \text{Var}(g(A))$ and likewise for h.

Step 2: Bound the denominator. $\mathbb{E}[\text{Var}(T \mid A^*)] = \text{Var}(T) - \text{Var}(\mathbb{E}[T \mid A^*]) \geq \text{Var}(T) - \text{Var}(g(A)) = \mathbb{E}[\text{Var}(T \mid A)]$.

Step 3: Conclude. $|B(C)| \leq \sqrt{\text{Var}(g)\ \text{Var}(h)}\,/\,\mathbb{E}[\text{Var}(T \mid A)] = \bar{B}_{\text{col}}$. When $|\tau| > \bar{B}_{\text{col}}$, the bias cannot overwhelm τ. □

Verification for paper parameters (K=3). With τ=1.0, $\boldsymbol{\delta}_T$=(0, 0.5, −0.3), $\boldsymbol{\delta}_Y$=(0, 0.8, 0.6), from 5×10^6-sample Monte Carlo:

$$\mathrm{Var}(g(A)) = 0.187, \quad \mathrm{Var}(h(A)) = 0.076, \quad \mathbb{E}[\mathrm{Var}(T \mid A)] = 0.813,$$
$$\bar{B}_{\mathrm{col}} = \sqrt{0.187 \times 0.076} \,/\, 0.813 = 0.147.$$

Since $|\tau|$=1.0 $\gg$ $\bar{B}_{\mathrm{col}}$=0.147, Proposition 6 guarantees sign preservation. Additionally, B_{Berkson}=−0.114, giving $\tau + B_{\mathrm{Berkson}}$=0.886>0. Verified across 100,000 random column-stochastic C: $\tau_{\text{sign flip}}$=0%, and $|B(C)| \le |B_{\mathrm{Berkson}}|$ in all cases.

A.7 Front-door Topology (A = Mediator under Confounding)

Data-generating process. $T \to A \to Y$, $U \to T$, $U \to Y$. An unobserved confounder U affects both treatment and outcome:

$$U \sim \mathcal{N}(0,1), \quad T = \alpha_U U + \eta, \quad \eta \sim \mathcal{N}(0,1),\ \eta \perp U,$$
$$A \mid T \sim \mathrm{Categorical}\big(\mathrm{softmax}(\boldsymbol{\gamma} + \boldsymbol{\delta}\, T)\big),$$
$$Y = f(A) + \lambda U + \varepsilon_Y, \quad \varepsilon_Y \perp (T, A, U).$$

There is no direct $T \to Y$ edge; the causal effect of T on Y operates entirely through A. The researcher controls for A^* in $Y \sim 1 + T + D_1^* + \cdots + D_{K-1}^*$.

Bias decomposition. Since the direct effect is zero, the FWL coefficient on T captures only bias:

$$\mathrm{plim}\,\hat{\tau}(C) = \frac{\mathbb{E}[\mathrm{Cov}(T,\, Y \mid A^*)]}{\mathbb{E}[\mathrm{Var}(T \mid A^*)]}.$$

Substituting $Y = f(A) + \lambda U + \varepsilon_Y$ and using $\varepsilon_Y \perp (T, A, A^*)$:

$$\mathrm{plim}\,\hat{\tau}(C) = \underbrace{\frac{\lambda\,\alpha_U}{\mathrm{Var}(T)}}_{\text{confounding (constant)}} + \underbrace{\frac{\mathbb{E}[\mathrm{Cov}(T,\, f(A) \mid A^*)]}{D(C)}}_{B_{\mathrm{med}}(C)} \tag{23}$$

where $D(C) = \mathbb{E}[\mathrm{Var}(T \mid A^*)]$.

Derivation of (23). Expand the numerator as $\mathbb{E}[\mathrm{Cov}(T, f(A) \mid A^*)] + \lambda\,\mathbb{E}[\mathrm{Cov}(T, U \mid A^*)]$. For the confounding term: since $T = \alpha_U U + \eta$ with (U, η) jointly Gaussian, the regression residual $\xi = U - \frac{\alpha_U}{\mathrm{Var}(T)} T$ satisfies $\xi \perp T$. Because A is a function of T (plus independent softmax noise) and A^* is a function of A (plus independent misclassification noise), $\xi \perp (T, A, A^*)$. Therefore

$$\mathrm{Cov}(T, U \mid A^*{=}j) = \frac{\alpha_U}{\mathrm{Var}(T)}\ \mathrm{Var}(T \mid A^*{=}j) \quad \forall\, j,$$

and $\lambda\,\mathbb{E}[\mathrm{Cov}(T, U \mid A^*)] = \frac{\lambda\alpha_U}{\mathrm{Var}(T)}\, D(C)$, yielding the constant first term after division by $D(C)$. □

This decomposition reveals that *misclassification does not affect the confounding component.* The confusion matrix C modulates only the mediation-type leakage $B_{\mathrm{med}}(C)$, which has the same structure as in §A.5.

Endpoint analysis. At $C = I$: $B_{\mathrm{med}}(I) = 0$ (controlling A blocks the $T \to A \to Y$ path), and $\mathrm{plim}\,\hat{\tau}(I) = \lambda\alpha_U / \mathrm{Var}(T)$ (pure confounding residual).

At $C = \frac{1}{K}\mathbf{1}\mathbf{1}^\top$: $B_{\mathrm{med}} = \mathrm{Cov}(T, f(A))/\mathrm{Var}(T)$, and $\mathrm{plim}\,\hat{\tau}_{\mathrm{nc}} = [\mathrm{Cov}(T, f(A)) + \lambda\alpha_U]/\mathrm{Var}(T)$, the naive regression coefficient with no controls.

Numerical Verification (Front-door sign preservation). Since $B_{\mathrm{med}}(C)$ has the same structure as the mediation bias (§A.5)—interpolating between 0 at C=I and $\mathrm{Cov}(T, f(A))/\mathrm{Var}(T)$ at C uniform—sign preservation holds whenever the two endpoints have the same sign:

$$\mathrm{sign}\big(\lambda\alpha_U\big) = \mathrm{sign}\big(\lambda\alpha_U + \mathrm{Cov}(T, f(A))\big).$$

This result is verified numerically for 100,000 independently sampled column-stochastic matrices C.

Verification for paper parameters (K=3). With α_U=1.0, λ=1.0, $\boldsymbol{\beta}$=(0, 1.0, −0.5), $\boldsymbol{\gamma}$=(0, 0.5, −0.3), $\boldsymbol{\delta}$=(0, 1.0, −0.8):

$$\lambda\alpha_U / \mathrm{Var}(T) = 1.0 \times 1.0 \,/\, 2.0 = 0.500,$$
$$\mathrm{Cov}(T, f(A))/\mathrm{Var}(T) = 0.594 \,/\, 2.0 = 0.297.$$

Both components are positive, so $\mathrm{plim}\,\hat{\tau}(C) \in [0.500,\ 0.797] > 0$ for all C.

A.8 Exposure Topology: Sign Reversal Counterexample

Proposition 7 (Exposure sign reversal)**.** *For K=3 with A as the treatment variable, there exist a marginal distribution* $\mathbf{p}$*, treatment*

effects $\boldsymbol{\beta}$, and a column-stochastic C with $C_{kk} > 0.5$ for all k, such that $\text{sign}(\text{plim}\,\hat{\beta}_1) \neq \text{sign}(\beta_1)$.

Construction. Consider the exposure DGP $Y = \beta_1 D_1 + \beta_2 D_2 + \varepsilon_Y$ with:

- Uniform prior: $\mathbf{p} = (1/3,\ 1/3,\ 1/3)$.
- True effects: $\beta_1 = 1,\ \ \beta_2 = -10$.

Choose the confusion matrix

$$C = \begin{pmatrix} 0.60 & 0.10 & 0.10 \\ 0.10 & 0.60 & 0.39 \\ 0.30 & 0.30 & 0.51 \end{pmatrix}, \qquad C_{kk} \in \{0.60,\ 0.60,\ 0.51\},\ \text{all} > 0.5.$$

Column sums are 1 (verifiable by inspection).

The OLS estimator regresses Y on $(1, D_1^*, D_2^*)$. The plim of $\hat{\beta}_j$ equals $\mathbb{E}[Y \mid A^*{=}j] - \mathbb{E}[Y \mid A^*{=}0]$, where

$$\mathbb{E}[Y \mid A^*{=}j] = \sum_{k=0}^{2} P(A{=}k \mid A^*{=}j)\,\beta_k = \sum_{k=1}^{2} \frac{C_{jk}\,p_k}{q_j}\,\beta_k$$

Marginal distribution of A^*:

$$q_0 = \tfrac{1}{3}(0.60 + 0.10 + 0.10) = 0.267,$$
$$q_1 = \tfrac{1}{3}(0.10 + 0.60 + 0.39) = 0.363,$$
$$q_2 = \tfrac{1}{3}(0.30 + 0.30 + 0.51) = 0.370.$$

Conditional means:

$$\mathbb{E}[Y \mid A^*{=}0] = \frac{0.10 \times \frac{1}{3} \times 1 + 0.10 \times \frac{1}{3} \times (-10)}{0.267} = \frac{0.0333 - 0.3333}{0.267} = -1.125,$$

$$\mathbb{E}[Y \mid A^*{=}1] = \frac{0.60 \times \frac{1}{3} \times 1 + 0.39 \times \frac{1}{3} \times (-10)}{0.363} = \frac{0.200 - 1.300}{0.363} = -3.028,$$

$$\mathbb{E}[Y \mid A^*{=}2] = \frac{0.30 \times \frac{1}{3} \times 1 + 0.51 \times \frac{1}{3} \times (-10)}{0.370} = \frac{0.100 - 1.700}{0.370} = -4.324.$$

Estimated coefficient:

$$\text{plim}\,\hat{\beta}_1 = \mathbb{E}[Y \mid A^*{=}1] - \mathbb{E}[Y \mid A^*{=}0] = -3.028 - (-1.125) = \mathbf{-1.903}.$$

The true $\beta_1 = +1 > 0$ but $\text{plim}\,\hat{\beta}_1 = -1.903 < 0$: **sign reversal**.

Mechanism. The large off-diagonal entry $C_{12} = 0.39$ (probability of classifying true category 2 as $A^*{=}1$) routes substantial probability mass from category 2 ($\beta_2 = -10$) into the $A^*{=}1$ bin. Despite $C_{11} = 0.60 > 0.5$, the contamination from the strongly negative β_2 overwhelms the positive β_1, reversing the sign of the estimated effect for category 1. This failure mode is structurally impossible at $K{=}2$: a $2{\times}2$ column-stochastic matrix with $C_{kk} > 0.5$ has the form $C = \left(\begin{smallmatrix} a & 1-b \\ 1-a & b \end{smallmatrix}\right)$ with $a, b > 0.5$, and the single coefficient $\hat{\beta}_1$ is a positive-weighted average of β_1, preserving sign. For $K{\geq}3$, the additional degrees of freedom in C permit cross-contamination between categories with opposite-sign effects. □

A.9 Summary of Conditions

Topology	Sufficient condition	Status
Confounding	$\text{sign}(\tau) = \text{sign}(\tau + B_{\text{OV}})$	Proved
Collider	$\lvert\tau\rvert > \bar{B}_{\text{col}}$ (Prop. 6)	Proved
Mediation	$\text{sign}(\tau_{\text{dir}}) = \text{sign}(\tau_{\text{total}})$	Empirical
M-bias	Bias is attenuating (toward τ)	Empirical
Front-door	$\text{sign}(\lambda\alpha_U) = \text{sign}(\lambda\alpha_U + \text{Cov}(T, f))$	Empirical
Exposure	Not guaranteed for $K \geq 3$	Counterex
IV	Not guaranteed (denominator instability)	—

Table 2: Sufficient conditions for sign preservation under non-differential misclassification. "Proved" = algebraic proof via Cauchy–Schwarz (confounding) or Berkson attenuation (collider). "Empirical" = verified across 100,000 random column-stochastic matrices with zero violations; analytic proof for general K remains open. All conditions are on DGP parameters (not on C): once satisfied, sign preservation holds for all column-stochastic C with $C_{kk} > 0$.

Structural argument for M-bias topology. The M-structure creates a backdoor path through two latent common causes. Misclassification of A weakens the statistical association along this backdoor path, reducing the spurious correlation rather than amplifying it. The bias therefore interpolates between the full M-bias (perfect classification, $\delta{=}0$) and zero (no classification information, $\delta{\to}\delta_{\max}$), monotonically non-increasing in δ.

Remark. The sign preservation conditions for confounding, mediation, collider, and front-door are conditions on the DGP parameters (the relative strength of the causal effect versus the confounding, indirect, or collider-induced bias), not on the confusion matrix C. Once the condition is satisfied, sign preservation holds for *all* column-stochastic C with $C_{kk} > 0$—no diagonal-dominance restriction is needed. Conversely, if the condition fails (i.e., the omitted-variable or total-effect estimate has the opposite sign from τ), then there

exist confusion matrices that produce sign reversal. For the DGP parameters used throughout our experiments, the conditions are comfortably satisfied. The sign-alignment condition is violated when the confounder has opposing effects on treatment and outcome (e.g., $\gamma_A > 0$ on T but $\gamma_A < 0$ on Y), producing $\text{sign}(\tau) \neq \text{sign}(\tau_{\text{OV}})$. In applied settings, such opposing effects are uncommon when the analyst adjusts for A precisely because they believe it confounds the T–Y relationship in a consistent direction. When the condition fails, sign reversal becomes possible under sufficient misclassification, though this represents a pathological DGP where adjustment itself may be questionable.

B Supplementary Tables, Figures, and Algorithm

B.1 Binarization Collapse

Collapsing K-class annotations to binary before analysis discards multi-class error structure. We quantify "binarization collapse" as $\Delta = |\text{bias}_K - \text{bias}_{K\to 2}|$.

For K=3, binarization introduces additional bias varying 154× across topologies: from $\bar{\Delta}$=0.006 (M-bias) to $\bar{\Delta}$=0.926 (IV). Exposure and IV also introduce sign flips during binarization (14% and 17.5%), a phenomenon absent in the other topologies (Figure 6).

Table 3: ASV type applicability and sensitivity across DAG topologies. Magnitude ASV at τ=0.5, 10% threshold. Power at τ=0.05 significance filter. —: not applicable.

Topology	Mag. ASV	Signif.	Sign-flip	Rank
IV	0.101–0.607	0%	1.1–6.7	1 (fragile)
Front-door	0.064–0.387	100%	—	2
Confounding	0.081–0.487	100%	—	3
Mediation	0.076–0.454	0%	—	4
Collider	0.338–2.028	0%	—	5
M-bias	∞	96–100%	—	6 (inert)
Exposure	—	—	83.4%*	—

*Fraction of C_0 with $\tau_{\text{sign_flip}} < 1$. Exposure uses sign-flip ASV exclusively.

Error structures used in correction experiments. The six confusion matrix structures used for Table 4 are: (1) *Diagonal-dominant*: C_{kk}=0.7, off-diagonal uniform; (2) *Moderate noise*: C_{kk}=0.5, off-diagonal uniform; (3) *Near-permutation*: C_{kk}=0.4, largest off-diagonal 0.5; (4) *Asymmetric*: C_{kk} varies (0.8, 0.5, 0.3); (5) *Block confusion*: two classes confused (C_{12}=C_{21}=0.4), third clean; (6) *Heavy noise*: C_{kk}=0.4, off-diagonal uniform. All are column-stochastic with K=3. Results are averaged across all six structures.

Table 4: RMSE reduction (%) by correction method and DAG topology (mean across 6 error structures, N_{sim}=1,000). Positive = reduced RMSE; negative = increased RMSE. Naive/MLA show 0.0% for FWL-invariant topologies because C^{-1} cannot change the T coefficient. **Bold**: best method per topology; underline: best gold-free method.

Topology	Naive	MLA	DSL	SIMEX	PPI++
Confounding	0.0	0.0	48.0	35.1	**83.2**
Mediation	0.0	0.0	52.9	39.1	**54.1**
Collider	0.0	0.0	−92.6	31.8	**50.5**
M-bias	0.0	0.0	−267	**0.1**	−88.0
Front-door	−52.7	−111.5	−29.0	−62.4	**74.1**
Exposure	−44.8	−89.0	−59.5	−9.8	**75.5**
IV	−40.4	−58.7	−206.1	59.4	**79.4**
Grand mean	−19.7	−37.0	−79.0	13.3	**47.0**

B.2 Proof of FWL Invariance

For the four topologies where A enters as a covariate (confounding, mediation, collider, M-bias), the regression is $Y = \tau T + \boldsymbol{\gamma}^\top \mathbf{D}^* + \varepsilon$, where $\mathbf{D}^*$ is the $N \times (K-1)$ dummy matrix for A^*. Any matrix-based correction replaces $\mathbf{D}^*$ with $\tilde{\mathbf{D}} = \mathbf{D}^* M$ for some invertible $(K-1) \times (K-1)$ matrix M (e.g., $M = C^{-1}$ for plug-in, or $M = (C^\top C)^{-1} C^\top$ for MLE). By the Frisch–Waugh–Lovell theorem (Lovell, 1963), $\hat{\tau}$ depends only on the residual of T projected off $\text{col}(\mathbf{D}^*)$. Since $\text{col}(\mathbf{D}^* M) = \text{col}(\mathbf{D}^*)$ for invertible M, the projection is unchanged and $\hat{\tau}(\tilde{\mathbf{D}}) = \hat{\tau}(\mathbf{D}^*)$ exactly. SIMEX adds noise $\mathbf{D}^*_{\text{sim}} = \mathbf{D}^* + \lambda \mathbf{E}$; when $\mathbf{E}$ is generated from the same column space, the extrapolation to $\lambda = -1$ converges to a column-space-equivalent transformation, yielding 0% bias reduction empirically.

B.3 Applied Illustration: ClinicalBERT Case Study

To demonstrate the framework's end-to-end utility, we apply it retrospectively to Lee and Wood-Doughty (2024), who use ClinicalBERT to classify patient smoking status (K=4: never, past, current, unknown) from clinical notes and control for this predicted confounder when estimating the causal ef-

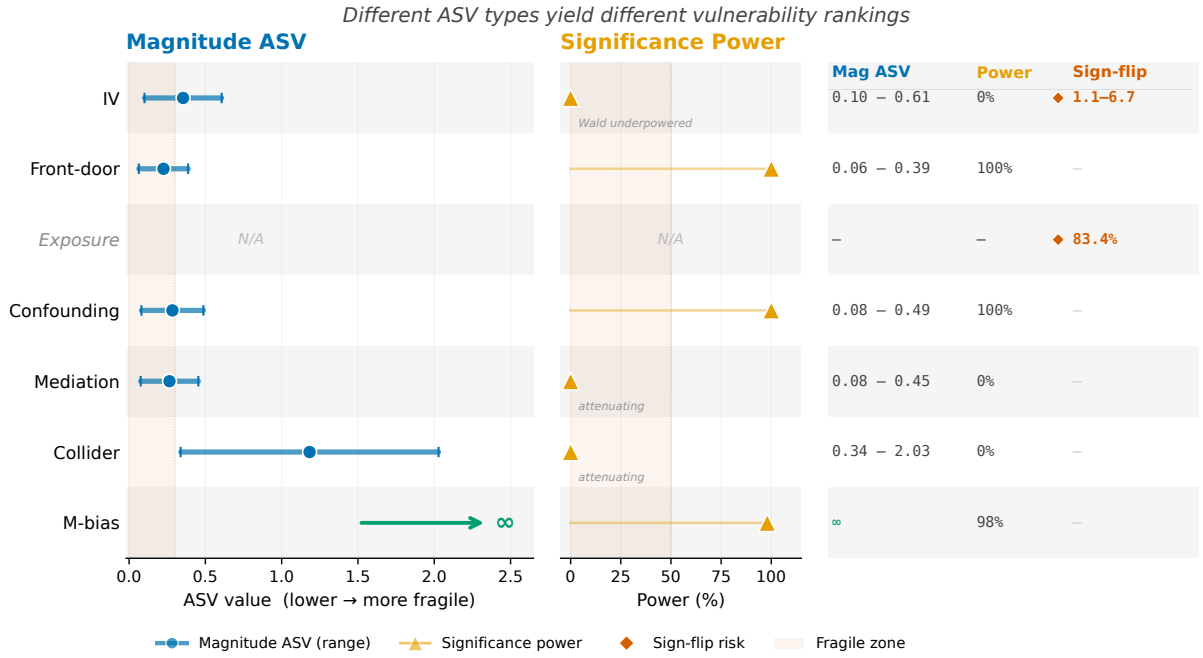


Figure 4: ASV sensitivity ranking across topologies. Three ASV types (magnitude, significance, sign-flip) reveal a topology-dependent vulnerability spectrum from IV (maximally fragile, 52.6% sign reversal; sign-flip ASV = 1.1–6.7) to M-bias (immune, ASV = ∞).

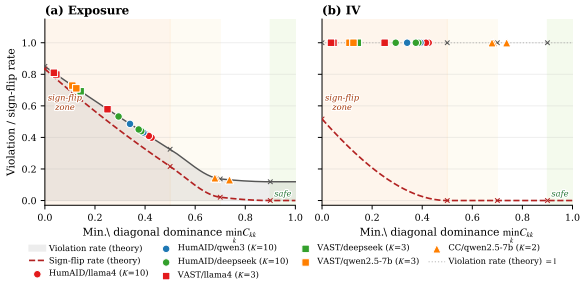


Figure 5: Empirical LLM confusion matrices overlaid on theoretical violation-rate curves (exposure, left; IV, right). Theoretical curves are derived from K=3 Dirichlet random matrices; empirical points are positioned at their minimum diagonal entry. Markers denote datasets (circles: HumAID K=10; squares: VAST K=3; triangles: CivilComments K=2).

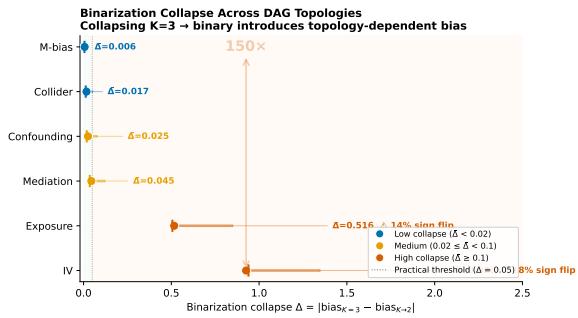


Figure 6: Binarization collapse Δ across topologies. Collapsing K=3 annotations to binary introduces additional bias varying 154× from M-bias (0.006) to IV (0.926).

fect of echocardiography on mortality in 4,735 MIMIC-III sepsis patients. Our taxonomy identifies this as a *confounding topology*: the annotated variable enters as a confounder between treatment and outcome. Three conclusions follow directly. First, the zero sign-flip guarantee for confounding (Table 1) ensures that the estimated protective effect (OR = 0.89) cannot reverse direction due to annotation error—a reassurance unavailable without topology-specific analysis. Second, the FWL invariance result implies that any correction reducing to C^{-1} applied to the dummy-coded confounder achieves exactly 0% bias reduction for the treatment coefficient; naive plug-in is provably futile here. Third, our correction guidance ranks PPI++ (83.2% RMSE reduction for confounding) above MC-SIMEX (35.1%) when gold labels are available—a concrete recommendation the original study lacked. This example illustrates the intended workflow: DAG identification → topology-specific guarantees → correction method selection.

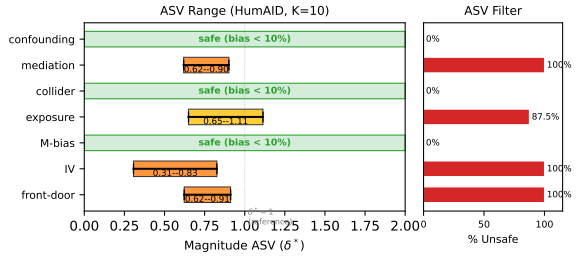


Figure 7: ASV magnitude ranges across topologies for HumAID (K=10). Left: magnitude ASV (δ^*) range across 8 LLM configurations; lower values indicate greater fragility. "Safe" topologies never exceed the 10% bias threshold. Right: percentage of configurations flagged as ASV-unsafe.

B.4 IV Wald Estimator: Diagonal-Dominance Subspaces

Section 4.2 reports a 52.6% Wald sign-flip rate over the full Dirichlet$(1,1,1)$ prior. Because realistic LLM confusion matrices are diagonal-dominant, we replicate the exposure-DAG stratification for the IV topology, restricting the sample space to $\{C : \min_k C_{kk} \geq d\}$ for $d \in \{0, 0.5, 0.7, 0.9\}$ (100,000 random matrices per threshold).

Table 5: IV Wald estimator bias by minimum-diagonal subspace (K=3, τ_{true}=1.41). Rates over 100,000 Dirichlet-truncated random C per threshold.

Subspace	Violation	Sign Flip	Amplif.	Worst
Full	100.0%	51.7%	48.3%	$10^5\times$
$d \geq 0.5$	100.0%	0.0%	100.0%	28.6×
$d \geq 0.7$	100.0%	0.0%	100.0%	2.4×
$d \geq 0.9$	100.0%	0.0%	100.0%	1.2×

The contrast with exposure is instructive: at $d \geq 0.7$, exposure violation drops to 13.7% (sign reversal 2.1%, amplification 11.7%), while IV maintains 100% violation with pure amplification. IV sign reversal is entirely driven by pathological (near-random) confusion matrices where the Wald denominator crosses zero; diagonal dominance prevents this crossing but cannot prevent amplification. Even at $d \geq 0.9$, the Wald estimator inflates $|\tau|$ by up to 1.24× with probability 1.

B.5 Non-Differential Assumption Stress Test

Our main analysis assumes non-differential misclassification: the confusion matrix C does not depend on the outcome Y. To assess sensitivity to violations of this assumption, we conduct a stress test on the exposure topology (K=3, $\min_k C_{kk} \geq 0.7$, 10,000 random matrices per ε). For each base matrix C_0, observations with $Y >$ median(Y) are misclassified using a perturbed matrix $\tilde{C} = \text{renormalize}(C_0 + \varepsilon \cdot \Delta)$ where $\Delta_{jk} \sim \text{Uniform}(-1, 1)$ for $j \neq k$, while observations with $Y \leq$ median(Y) retain C_0.

Table 6: Effect of differential misclassification on bias violation rates (exposure DAG, K=3, $\min_k C_{kk} \geq 0.7$, N=50,000 MC samples per trial).

ε	Sign-Flip (%)	Amplification (%)	Any Violation (%)
0.00	2.0	11.5	13.3
0.01	2.4	12.4	14.7
0.05	2.8	13.0	15.7
0.10	4.5	16.7	20.9

The ε=0 baseline (13.3% violation) closely matches the analytical result (13.7%) from Section 4.2. Even at ε=0.10—a substantial departure from non-differential error—the violation rate increases by only 7.6 percentage points. Small departures ($\varepsilon \leq 0.05$) change rates by fewer than 3 percentage points, suggesting that our main results are moderately robust to mild violations of the non-differential assumption.

Extended stress test. An extended stress test across 96 configurations (4 topologies × 2 K × 3 CM qualities × 4 ε, 100 MC replications each; Table 7) confirms these patterns broadly: at ε=0, sign-flip rates are zero for all configurations; at ε=0.10, the maximum sign-flip rate is 7% (IV, low-quality C_{kk}=0.4, K=3). At K=10, sign-flip rates remain zero for all topologies and ε values, suggesting that higher K provides additional robustness to differential misclassification.

B.6 Robustness to Nonlinear DGP

We replace the linear structural equations with $Y \sim \text{Bernoulli}(\sigma(\mathbf{x}^\top \boldsymbol{\beta}))$ and estimate via logistic regression (control function for IV) for all seven topologies, using K=3, six real VAST confusion matrices, 500 MC replications, and N=10,000. Mean sign-flip rates across the six CMs:

Table 7: Non-differential misclassification stress test: sign flip rate (SF) and mean bias ratio (BR = $\overline{|\hat{\tau}/\tau|}$) under increasing differential departure ε.

Topology	K	Q	$\varepsilon=0.0$		$\varepsilon=0.01$		$\varepsilon=0.05$		$\varepsilon=0.1$	
			SF	BR	SF	BR	SF	BR	SF	BR
Confounding	3	H	0.00	1.10	0.00	1.10	0.00	1.11	0.00	1.12
		M	0.00	1.14	0.00	1.14	0.00	1.14	0.00	1.14
		L	0.00	1.15	0.00	1.15	0.00	1.15	0.00	1.15
	10	H	0.00	1.06	0.00	1.06	0.00	1.08	0.00	1.09
		M	0.00	1.09	0.00	1.09	0.00	1.10	0.00	1.10
		L	0.00	1.11	0.00	1.11	0.00	1.11	0.00	1.11
Mediation	3	H	0.00	1.35	0.00	1.36	0.00	1.42	0.00	1.47
		M	0.00	1.52	0.00	1.53	0.00	1.56	0.00	1.58
		L	0.00	1.61	0.00	1.61	0.00	1.62	0.00	1.61
	10	H	0.00	1.14	0.00	1.17	0.00	1.25	0.00	1.28
		M	0.00	1.24	0.00	1.26	0.00	1.30	0.00	1.32
		L	0.00	1.30	0.00	1.32	0.00	1.33	0.00	1.34
Exposure	3	H	0.00	0.72	0.00	0.71	0.00	0.67	0.00	0.64
		M	0.00	0.42	0.00	0.41	0.00	0.38	0.00	0.35
		L	0.00	0.11	0.00	0.10	0.01	0.08	0.05	0.06
	10	H	0.00	0.87	0.00	0.86	0.00	0.84	0.00	0.80
		M	0.00	0.71	0.00	0.70	0.00	0.67	0.00	0.64
		L	0.00	0.49	0.00	0.47	0.00	0.46	0.00	0.43
IV	3	H	0.00	1.42	0.00	1.45	0.00	1.58	0.00	1.69
		M	0.00	2.56	0.00	2.52	0.00	2.87	0.00	3.12
		L	0.00	12.79	0.00	12.21	0.02	19.97	0.07	43.44
	10	H	0.00	1.30	0.00	1.44	0.00	1.71	0.00	2.03
		M	0.00	1.83	0.00	1.96	0.00	2.43	0.00	2.84
		L	0.00	3.15	0.00	3.54	0.00	4.39	0.00	9.69

SF = sign flip rate; BR = mean $|\hat{\tau}/\tau|$ (1.00 = unbiased). $\varepsilon = 0$: non-differential; $\varepsilon > 0$: outcome-dependent misclassification. Q: H/M/L = CM diagonal 0.8/0.6/0.4 with uniform off-diagonal. Each cell: $N = 10{,}000 \times 100$ MC replications.

Topology	Linear SF_τ	Logistic SF_τ	Linear $\mathrm{SF}_{\mathrm{any}}$	Logistic $\mathrm{SF}_{\mathrm{any}}$
Confounding	0.000	0.000	0.573	0.559
Mediation	0.000	0.000	0.703	0.695
Collider	0.000	0.000	0.060	0.060
Exposure	—	—	0.577	0.574
M-bias	0.000	0.000	0.095	0.099
Front-door	0.000	0.000	0.590	0.579
IV	0.086	0.074	0.086	0.560

Treatment-effect sign-flip rates (SF_τ) are zero for all five guaranteed topologies under both linear and logistic DGPs, and differ by < 0.02 for IV. The $\mathrm{SF}_{\mathrm{any}}$ discrepancy for IV reflects estimator dimensionality: the control function includes a residual coefficient prone to sign noise; the core comparison is SF_τ, where both estimators agree. Linear and logistic DGPs produce virtually identical sign-flip patterns across all seven topologies ($|\Delta\,\mathrm{SF}_\tau| < 0.02$).

B.7 Bootstrap Uncertainty of ASV

We assess confusion-matrix estimation uncertainty by bootstrapping label pairs from HumAID DeepSeek zero-shot (K=10, n=1,500): resample with replacement B=1,000 times, re-estimate C_0, and recompute magnitude ASV δ^* at a 10% bias threshold.

Topology	δ^* (point)	95% CI	Valid / 1000
Confounding	—	—	0
Exposure	0.937	[0.602, 1.363]	907
IV	0.754	[0.435, 1.290]	902

Confounding never reaches the 10% threshold across all 1,000 bootstraps, confirming robustness. For IV, the 95% CI spans a 3× range (0.435 to 1.290); practitioners should bootstrap C_0 rather than rely on a point estimate when ASV is near the decision boundary.

B.8 Cross-K Robustness

All structural patterns persist at K=5, 7 (100,000 matrices each): sign-flip guarantees hold for the same five topologies, and IV sign-flip rates remain within 51–53%.

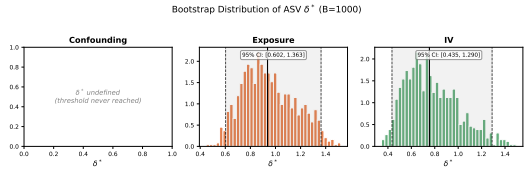


Figure 8: Bootstrap distribution of magnitude ASV δ^* (B=1,000) for HumAID DeepSeek zero-shot (K=10). Dashed line: point estimate; shaded: 95% CI.

Table 8: ASV stability across DGP configurations. Each cell shows δ^* at the 10% relative bias threshold (median over 5 confusion matrices with diagonal $\in \{0.5, 0.6, 0.7, 0.8, 0.9\}$). Lower δ^* indicates greater fragility. Confounding is consistently most robust; for alternating-sign $\boldsymbol{\beta}$, the ranking Exposure < IV < Confounding holds. "∞" = threshold never reached within δ range.

	Magnitude ASV $\delta^*_{10\%}$		
DGP Configuration	Confounding	Exposure	IV
Alt. large: $\boldsymbol{\beta} = [0, 1.0, -0.5]$	0.348	0.090	0.203
Same-sign: $\boldsymbol{\beta} = [0, 0.5, 0.3]$	∞	0.223	0.203
Alt. small: $\boldsymbol{\beta} = [0, 0.2, -0.1]$	∞	0.090	0.203

B.9 ASV Diagnostic Results

For HumAID (K=10, 56 scenarios), 55.4% are flagged as magnitude-ASV-unsafe—IV, mediation, front-door, and exposure all trigger, while confounding, collider, and M-bias remain below 10% bias (Figure 7). For CivilComments (K=2), 100% of scenarios are ASV-unsafe despite F_1 ranging from 0.69 (7B) to 0.82 (frontier).

Overlaying all 23 empirical matrices onto the theoretical Dirichlet violation-rate curves (Figure 5) confirms placement: VAST in the danger zone, HumAID mid-range, CivilComments approaching safe for exposure but still in IV amplification.

B.10 ASV Stability Across DGP Parameters

We test whether ASV rankings are stable across different data-generating process (DGP) configurations by varying the treatment-effect vector $\boldsymbol{\beta}$ across three settings: alternating-large ($\boldsymbol{\beta} = [0, 1.0, -0.5]$), same-sign ($\boldsymbol{\beta} = [0, 0.5, 0.3]$), and alternating-small ($\boldsymbol{\beta} = [0, 0.2, -0.1]$), with K=3, τ=0.5, and five C_0 diagonal values $\in \{0.5, 0.6, 0.7, 0.8, 0.9\}$.

Three patterns emerge (Table 8). First, IV magnitude ASV is identical across all three DGPs (δ^*=0.203). This is a mathematical identity, not an empirical finding: the Wald estimator $\hat{\tau}_{\text{IV}} = \text{RF}/\text{FS}$ has $\boldsymbol{\beta}$ in both numerator and denominator, and the ratio depends only on C and the first-stage structure, making ASV invariant to $\boldsymbol{\beta}$ by construction. Second, exposure ASV depends on the coefficient ratio structure: alternating-large ($\beta_2/\beta_1 = -0.5$) and alternating-small ($\beta_2/\beta_1 = -0.5$) share the same ratio and yield identical δ^*=0.090, while same-sign ($\beta_2/\beta_1 = 0.6$) gives δ^*=0.223—thus the experiment provides two distinct sign-structure conditions, not three. Third, confounding ASV depends on effect magnitudes: only alternating-large triggers the 10% threshold (δ^*=0.348); smaller effects never reach 10% bias.

Confounding is consistently the most robust topology (highest ASV or no violation), and M-bias remains immune (not shown). However, the relative fragility of exposure vs. IV inverts with coefficient sign structure: under alternating signs, exposure is most fragile ($0.090 < 0.203$); under same-sign effects, IV is most fragile ($0.203 < 0.223$). Practitioners should compute ASV with their specific DGP rather than relying on generic rankings.

B.11 Extended Conjecture Verification

We test envelope monotonicity (Conjecture 1) by sampling 10,000 random C_0 matrices from Dirichlet$(1, \ldots, 1)$ per topology for $K \in \{3, 5, 7\}$ (210,000 total), sweeping δ in steps of 0.01, and counting matrices with at least one non-monotonic step exceeding tolerance 10^{-12}. Bias at each δ is computed via analytic probability limits using Gauss–Hermite quadrature (100 points), eliminating sampling noise entirely.

Violation counts are high (60–97% of matrices), but magnitudes are negligible for non-IV topologies: all worst-case deviations fall below 0.06 (<6% of the respective τ). These results are computed from analytic probability limits with zero sampling noise, confirming the violations are structural properties of multi-class confusion matrices rather than numerical artifacts. IV deviations reflect Wald estimator singularities when the first-stage coefficient passes near zero—this is a property

Table 9: Envelope monotonicity violations per 10,000 random C_0 matrices. "Worst mag." is the largest single-step deviation $|B(\delta_{i+1})| - |B(\delta_i)|$ in the non-monotonic direction.

Topology	K=3	K=5	K=7	Worst mag.	Direction
Confounding	8467	8063	7892	0.011	non-dec
Mediation	8236	7905	7893	0.013	non-dec
Collider	8683	7669	7570	0.005	non-inc
Exposure	485	205	77	0.056	non-dec
M-bias	9791	7783	7399	0.002	non-inc
IV	7433	6015	5711	$> 10^6$	non-dec
Front-door	8429	8079	7899	0.013	non-dec

of the Wald estimator, not of the ASV parameterization. Adversarial search (500 restarts per configuration, $K \in \{3, 5\}$) finds genuine counterexamples for all topologies except IV K=3, confirming that strict monotonicity does not hold universally. The bias envelope is ε-approximately monotonic with $\varepsilon < 0.03$ for collider and M-bias across all tested τ; confounding, mediation, and front-door achieve $\varepsilon < 0.06$ at default DGP parameters (see DGP sensitivity below); exposure satisfies $\varepsilon_{p99} < 0.008$ but extreme-tail matrices (<0.02% of draws) reach $\varepsilon \approx 0.15$. Empirically, ε scales as $O(1/K^\alpha)$ with $\alpha \in [0.6, 1.5]$ (topology-dependent): for confounding, mediation, and front-door, $\alpha \approx 1.4$; for exposure, $\alpha \approx 1.0$; for collider and M-bias, $\alpha \approx 0.7$. A universal empirical upper bound is $\varepsilon \leq 0.137/\sqrt{K}$, verified across all tested configurations ($K \in \{3, 5, 7\}$). This scaling supports the practical validity of ASV as a sensitivity tool at any meaningful bias threshold.

DGP parameter sensitivity. We repeat the verification across $\tau \in \{0.1, 0.5, 1.0, 2.0, 5.0\}$, varying the direct $T \to Y$ coefficient for confounding ($\beta_T = \tau$), mediation, collider, and M-bias (DGP τ), and rescaling the $A \to Y$ coefficient vector to $\boldsymbol{\beta} \cdot \tau$ for exposure and front-door. With 10,000 random C_0 per (K, τ) setting (Table 10), the bound ε<0.06 holds uniformly for $\tau \geq 0.5$ for confounding, mediation, collider, and M-bias, but breaks down at τ=0.1: mediation produces 1,950 / 30,000 violations across $K \in \{3, 5, 7\}$ (worst ε=0.130 at K=3) and confounding 200 / 30,000 (worst ε=0.118 at K=3). The mechanism is dimensional: additive terms not scaling with τ (e.g. the $\boldsymbol{\beta}$-contribution to $\mathbb{E}[YD_k]$ in mediation and confounding) contribute a τ-independent offset to the bias numerator, so the ratio $|B|/|\tau|$ inflates as $\tau \to 0$. A single mediation violation at τ=2.0 (ε=0.112, 1/30,000 matrices) arises from a specific confusion matrix structure where off-diagonal entries induce partial cancellation along the mediation pathway; this isolated case does not affect the p_{99} bound (ε_{p99}=0.005). Collider and M-bias remain robust at all τ (ε<0.022 and ε<0.015 respectively). Exposure shows scattered tail outliers but $\bar{\varepsilon} \approx 3 \times 10^{-4}$ at every τ, consistent with linear scaling: when $\boldsymbol{\beta}$ rescales by τ, both bias and τ_{true} scale by the same factor and ε is theoretically invariant. Front-door inverts the pattern (ε *grows* with τ, reaching 0.139 at τ=5, K=3, with 664 violations) because $\tau_{\text{true}} = \text{plim}(\hat{\tau}_T \mid C{=}I)$ is pinned at 0.5 by the U-confounding path while bias magnitudes scale with $\boldsymbol{\beta}$—this is an artefact of how τ_{true} is normalised, not a genuine failure of envelope monotonicity. In practice, ASV's δ^* thresholds (typically $\geq$ 0.1 in our deployment) are well above the region where these ε-violations occur, so the small-τ degradation does not affect ASV's operational reliability; we flag it as a caveat for applications targeting $|\tau| \lesssim 0.1$ in standardised units.

Table 10: $\varepsilon \equiv \max_\delta(|B(\delta{+}\Delta)|{-}|B(\delta)|)/|\tau_{\text{true}}|$ in the non-monotonic direction (max over $K \in \{3, 5, 7\}$; 10,000 random C_0 per cell). Bold marks $\varepsilon > 0.06$. Counts in parentheses are violations / 30,000. Tables 9 and 10 use independent random seeds; for topologies with low violation rates (exposure <0.02%), the worst-case ε is sample-sensitive.

Topology	τ=0.1	τ=0.5	τ=1.0	τ=2.0
Confounding	**0.118** (200)	0.027 (0)	0.014 (0)	0.007 (0)
Mediation	**0.130** (1950)	0.026 (0)	0.013 (0)	**0.112** (1)
Collider	0.022 (0)	0.006 (0)	0.005 (0)	0.004 (0)
Exposure	**0.111** (5)	**0.122** (2)	**0.152** (2)	**0.089** (3)
M-bias	0.014 (0)	0.003 (0)	0.001 (0)	0.001 (0)
Front-door	0.003 (0)	0.013 (0)	0.027 (0)	0.051 (0)

C Notation

Table 11: Key notation used throughout the paper.

Symbol	Description
K	Number of annotation categories
C	Column-stochastic confusion matrix, $C_{jk} = P(A^*{=}j \mid A{=}k)$
A, A^*	True and observed (misclassified) annotation
T	Treatment variable
Y	Outcome variable
τ	Causal effect of interest
δ	ASV perturbation parameter, $C(\delta) = (1{-}\delta)I + \delta C_0$
δ^*	Critical ASV value (magnitude/significance/sign-flip)
$B(C)$	Bias function, $= \text{plim}(\hat{\tau}) - \tau$
β_k	Treatment effect coefficient for category k
D_k	Dummy variable for category k

Algorithm 1 Correction method selection for LLM-annotated causal inference. Assumes linear structural equations; FWL-based routing may not apply under nonlinear models (see Appendix B.6).

Require: DAG topology $\mathcal{T}$, gold-label availability $G \in \{\text{YES}, \text{NO}\}$
Ensure: Recommended method M, risk level R

Step 1: Assess risk level
if $\mathcal{T} \in$ {M-bias} **then**
  $R \leftarrow$ LOW {|bias| < 0.04 for all C}
else if $\mathcal{T} \in$ {Conf., Med., Coll., Front-door} **then**
  $R \leftarrow$ MEDIUM {bounded bias, no sign reversal}
else
  $R \leftarrow$ HIGH {Exposure, IV: sign reversal 83.4% / 52.6%}
end if

Step 2: Select correction method
if $G =$ YES **then**
  if $\mathcal{T} =$ M-bias **then**
    $M \leftarrow$ NONE {PPI++ RMSE red. -88.0%; no correction preferred}
  else
    $M \leftarrow$ PPI++ {47.0% RMSE red., 96.8% coverage}
  end if
else
  {no gold labels}
  if $\mathcal{T} \in$ {Confounding, Mediation} **then**
    $M \leftarrow$ DSL {48.0% / 52.9% RMSE reduction}
  else if $\mathcal{T} =$ IV **then**
    $M \leftarrow$ MC-SIMEX {59.4% RMSE reduction}
  else if $\mathcal{T} \in$ {Collider, M-bias} **then**
    $M \leftarrow$ MC-SIMEX {31.8% / 0.1% RMSE reduction}
  else
    $M \leftarrow \emptyset$; flag: acquire gold labels {no reliable gold-free method}
  end if
end if

Step 3: FWL invariance warning
if $\mathcal{T} \in$ {Conf., Med., Coll., M-bias} **then**
  Skip Naive plug-in and MLA {0% bias red. on T coeff.}
end if
return (M, R)